



08. Thoracic cavity, mediastinum. Chambers of the heart, external features. Structure of heart wall, valves, fibrous skeleton. Pericardium



**Semmelweis University,
Faculty of Medicine,
Department of Anatomy, Histology and Embryology
Macroscopic Anatomy and Embryology II.**

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Mediastinum

anterior mediastinum - cardiac, supracardiac

posterior)mediastinum

Circulatory system, position of the heart

Surfaces of the heart

Cavities of the heart:

Right atrium

Right ventricle

Left atrium

Left ventricle

Cardiac skeleton, anulus fibrosus

Valves of the heart

Cuspid valves

Semilunar valves

Heart wall, layers

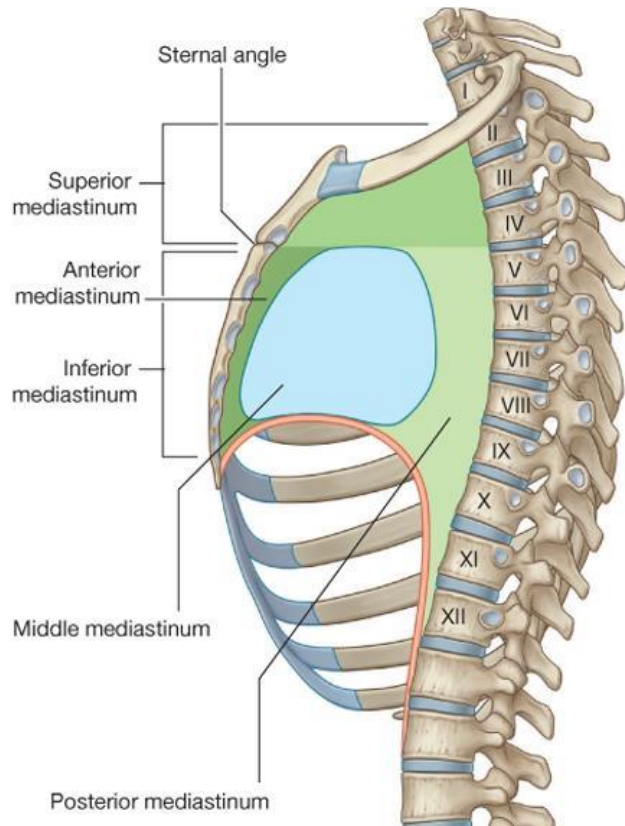
Interventricular septum

Pericardium

Mediastinum

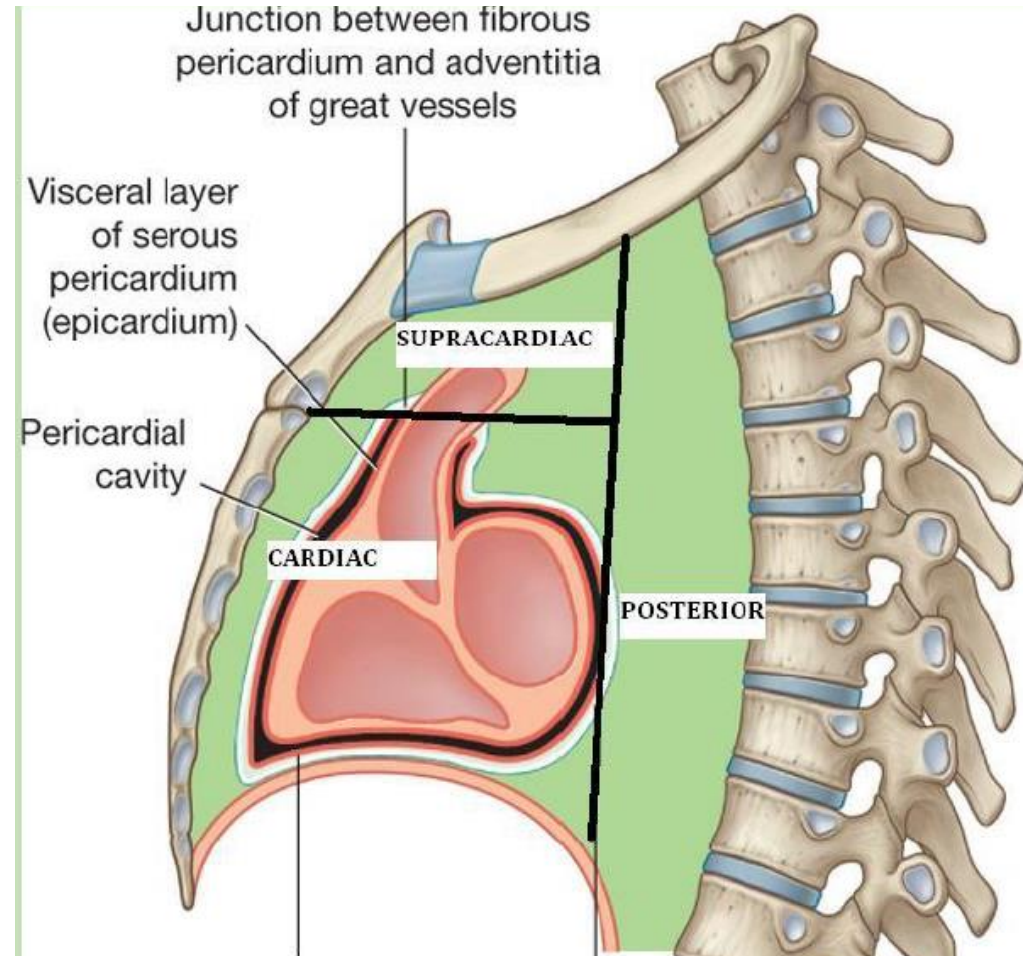
Subdivisions of mediastinum - versions

MEDIASTINUM: space in the thoracic cavity, above the diaphragm, between the two lungs



Version 1

Superior / Inferior (ant. + med. + post)



Version 2

Anterior (cardiac + supracardiac) / Posterior

Layers in the **supracardiac mediastinum**

1. Thymus
2. Large veins (SVC, Brachiocephalic, Azygos)
3. Phrenic n., pericardiophrenic a.v.
4. Aortic arch + branches, vagus n. (recurrent lar.)
5. Trachea

Structures in the **cardiac mediastinum**

pericardium (layers), heart, ascending aorta,
pulmonary trunk, superior vena cava
phrenic nerve, pericardiophrenic a.v.

Structures in the **posterior mediastinum**

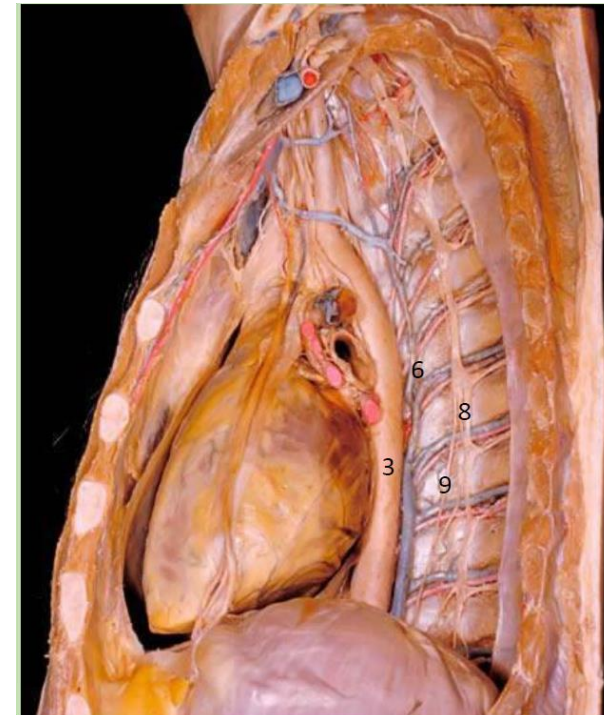
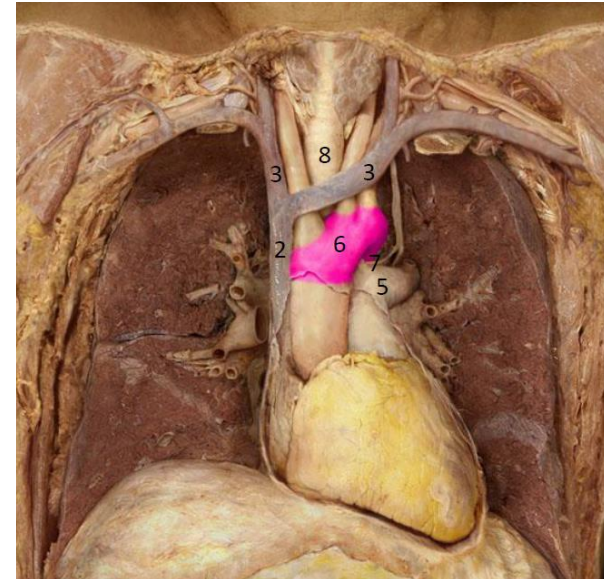
Esophagus

Thoracic aorta and branches (esophageal, tracheal,
bronchial, pericardial, sup.phrenic a., post.intercostal a.)

Azygos - hemiazygos, acc. hemiazygos veins

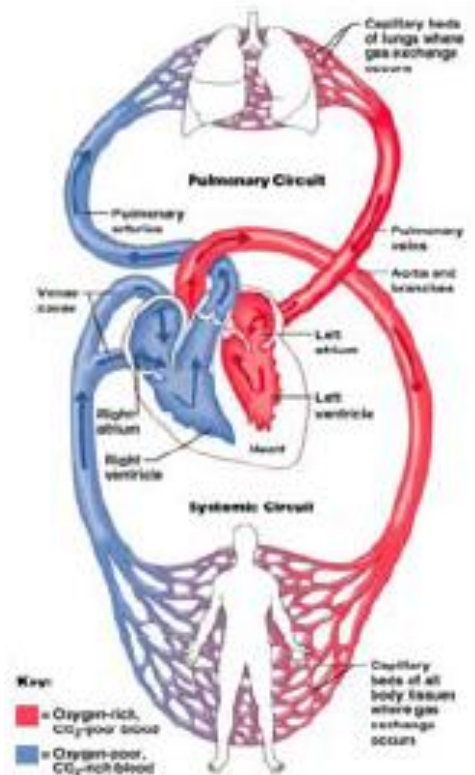
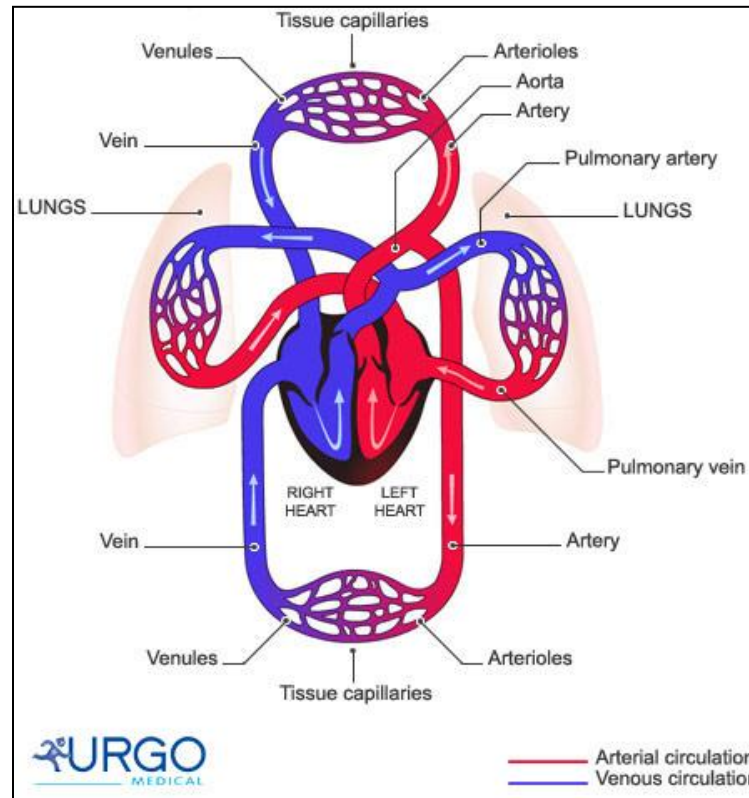
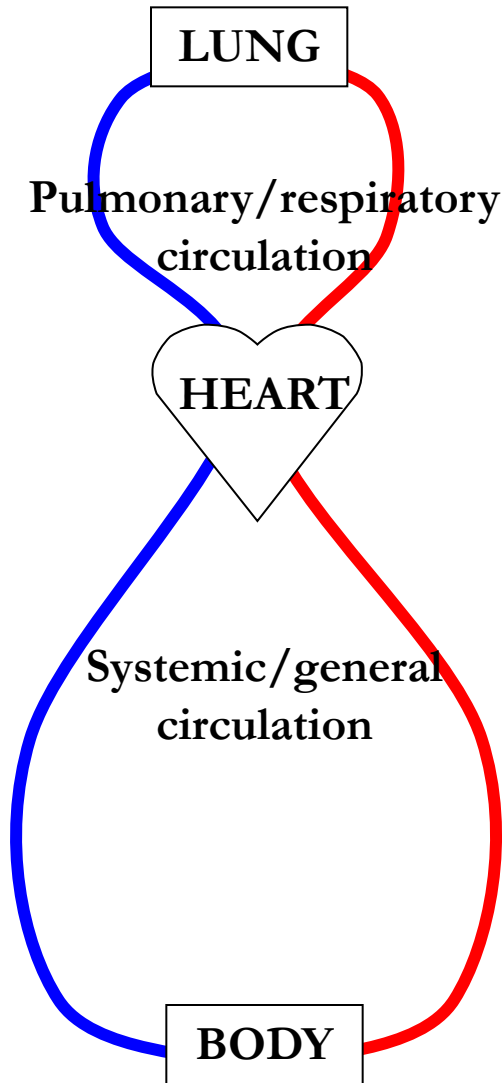
Thoracic duct

Nerves: vagus n., intercostal nerves, sympathetic
trunk, greater and lesser splanchnic nerves)

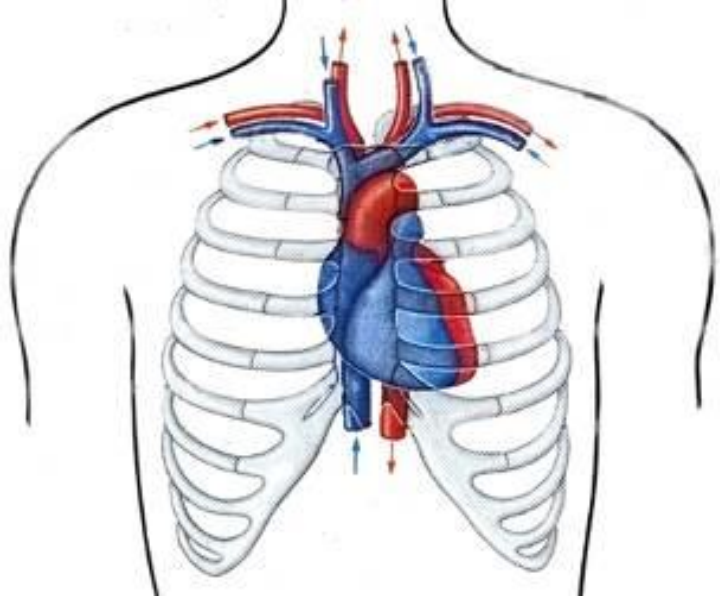


Circulatory system position of the heart

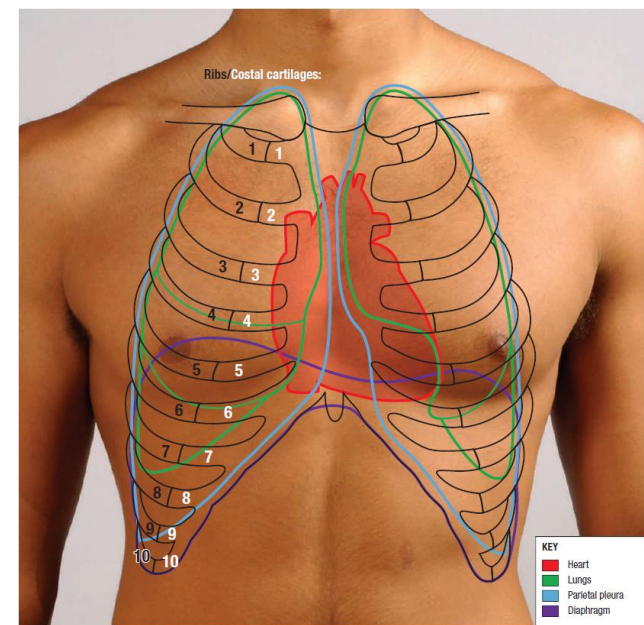
The circulatory system



Position of heart

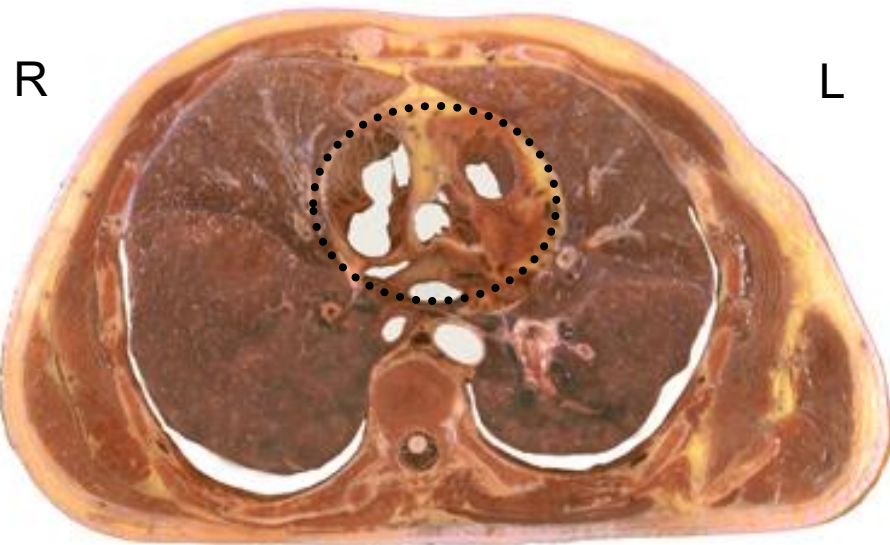
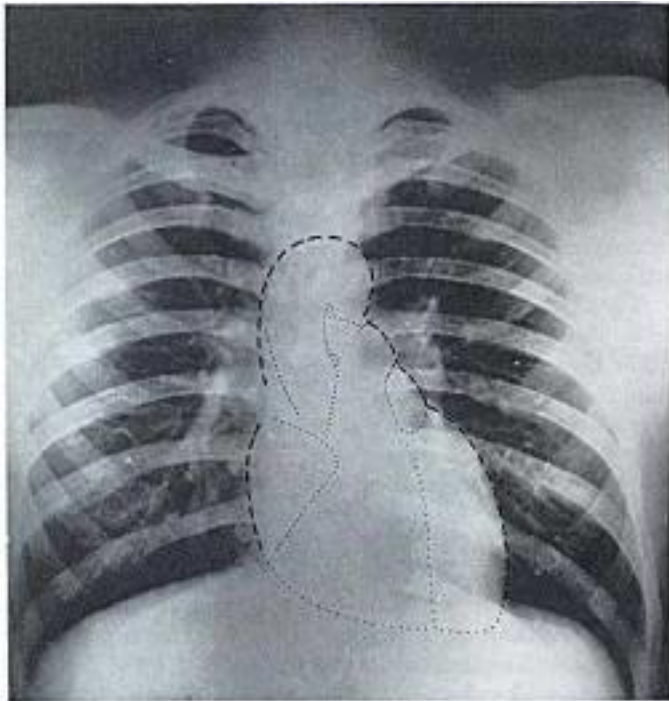


Position of heart



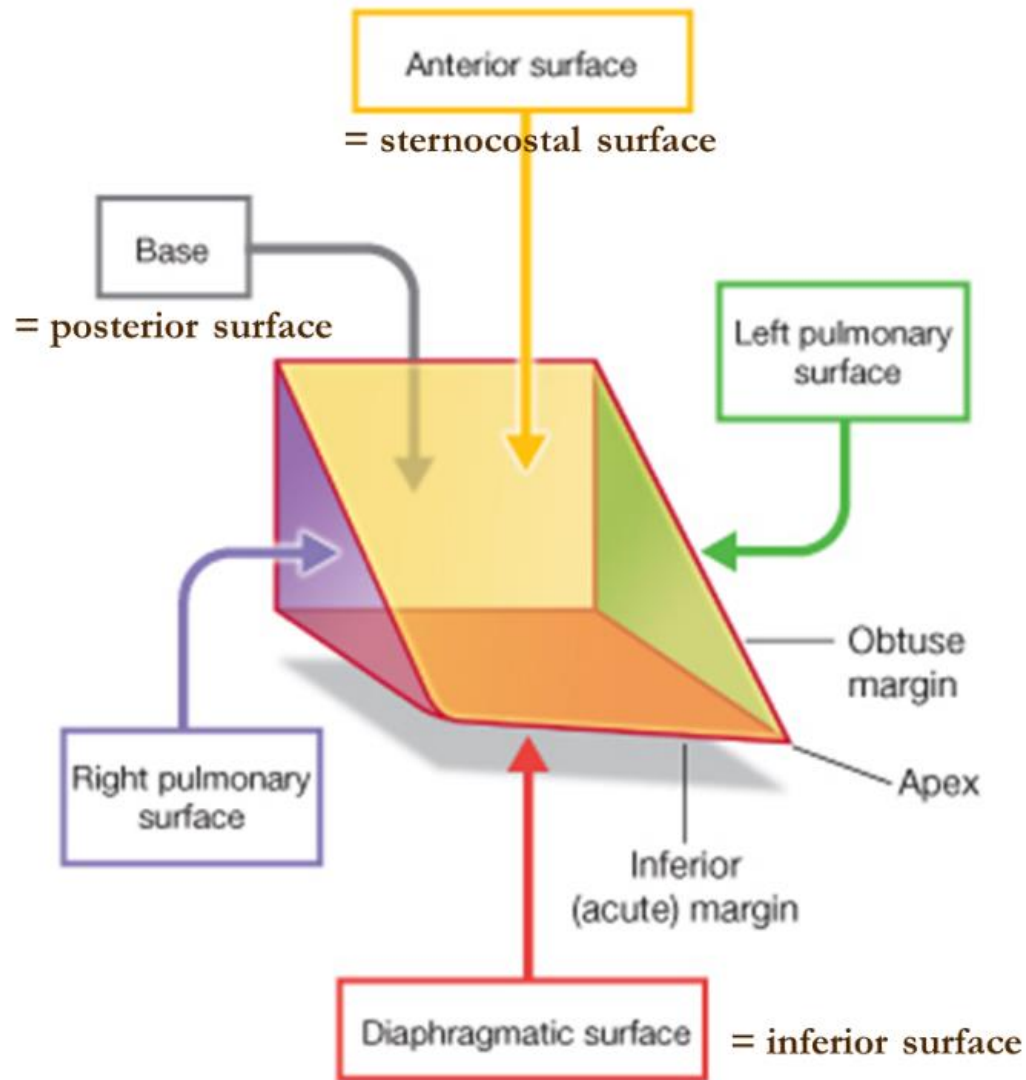
Thoracic cavity

- mediastinum cardiacum
- above diaphragm
- between mediastinal pleuras

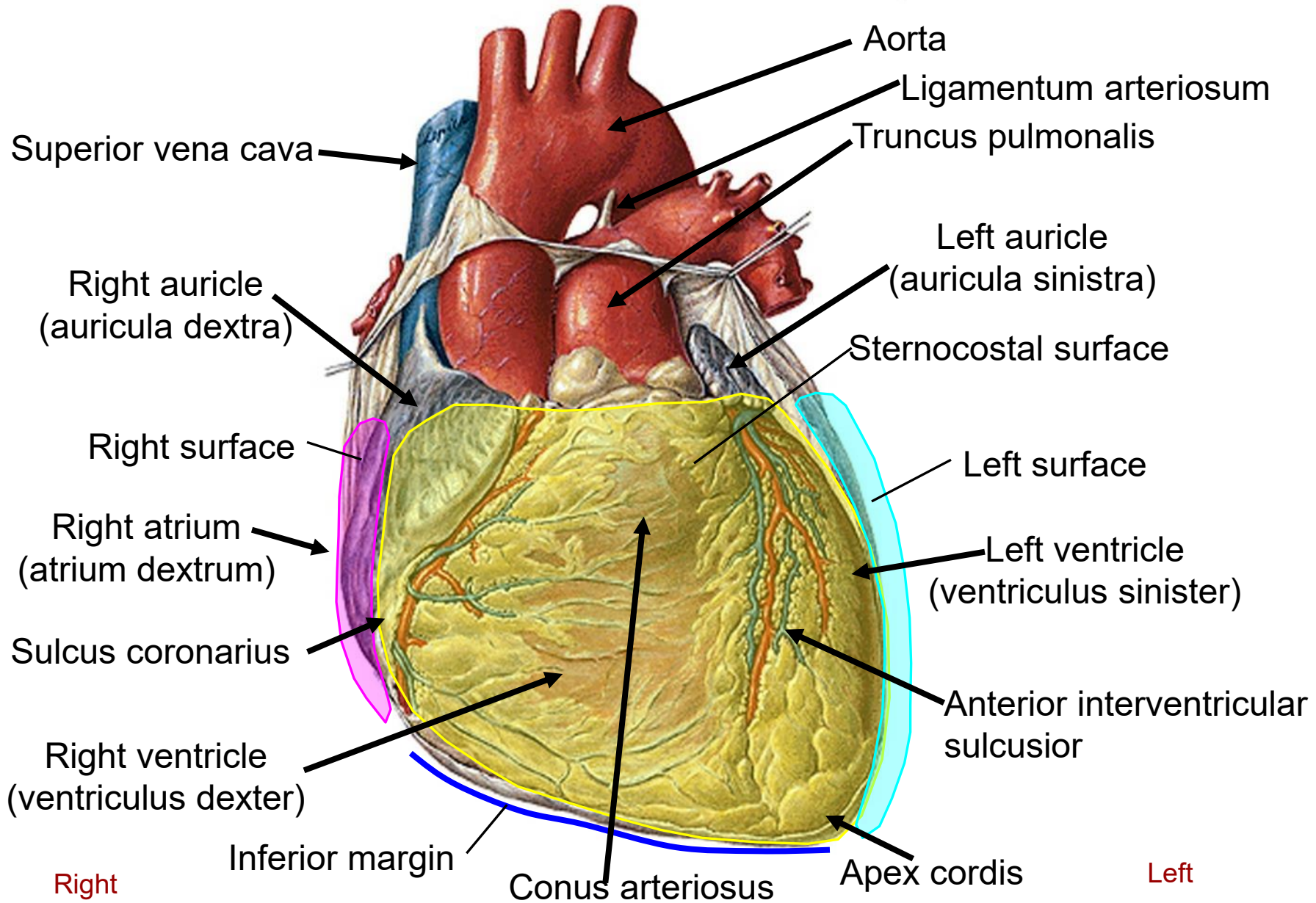


Surfaces of the heart

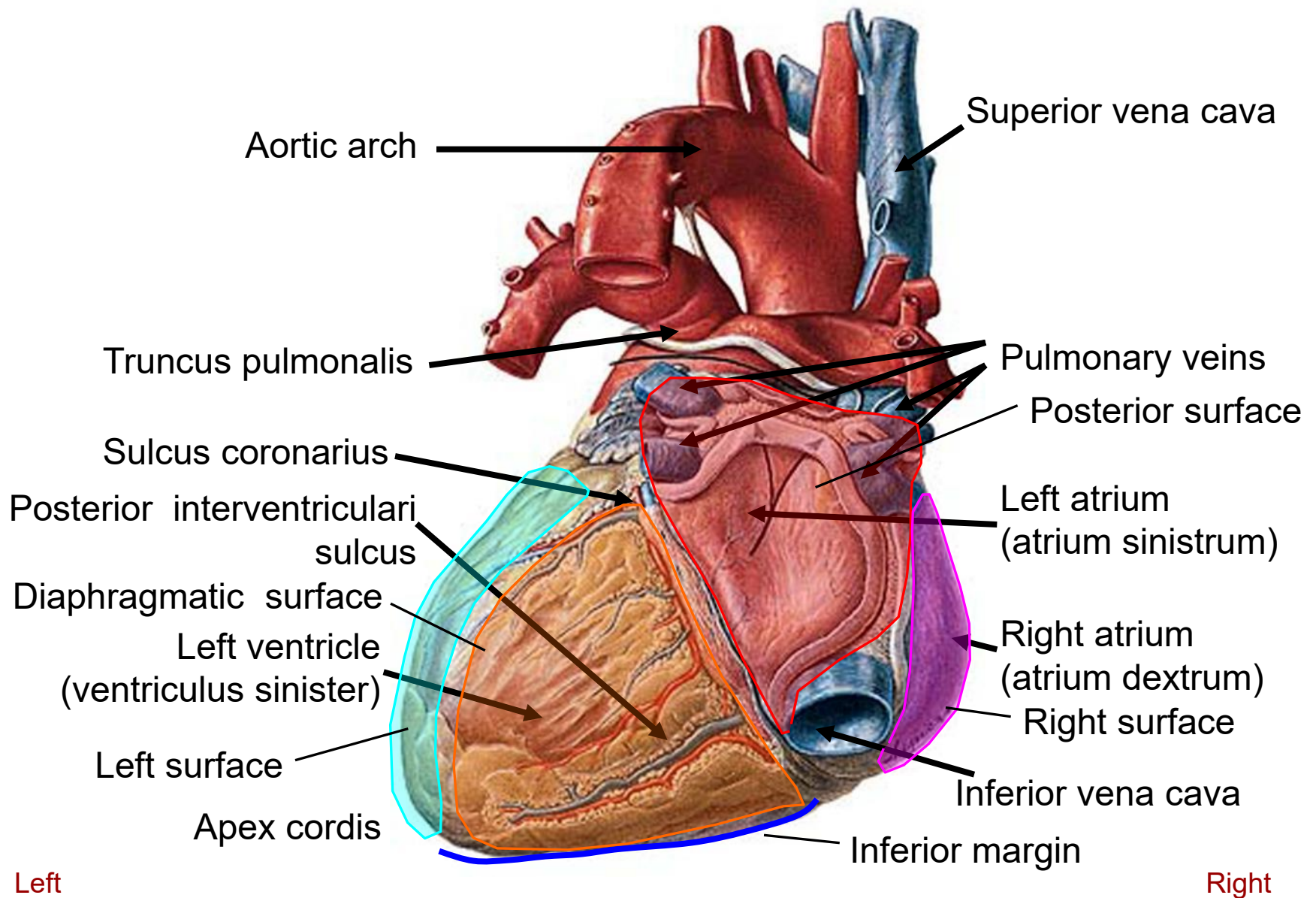
Surfaces of heart



Heart: sternocostal surface (facies sternocostalis)

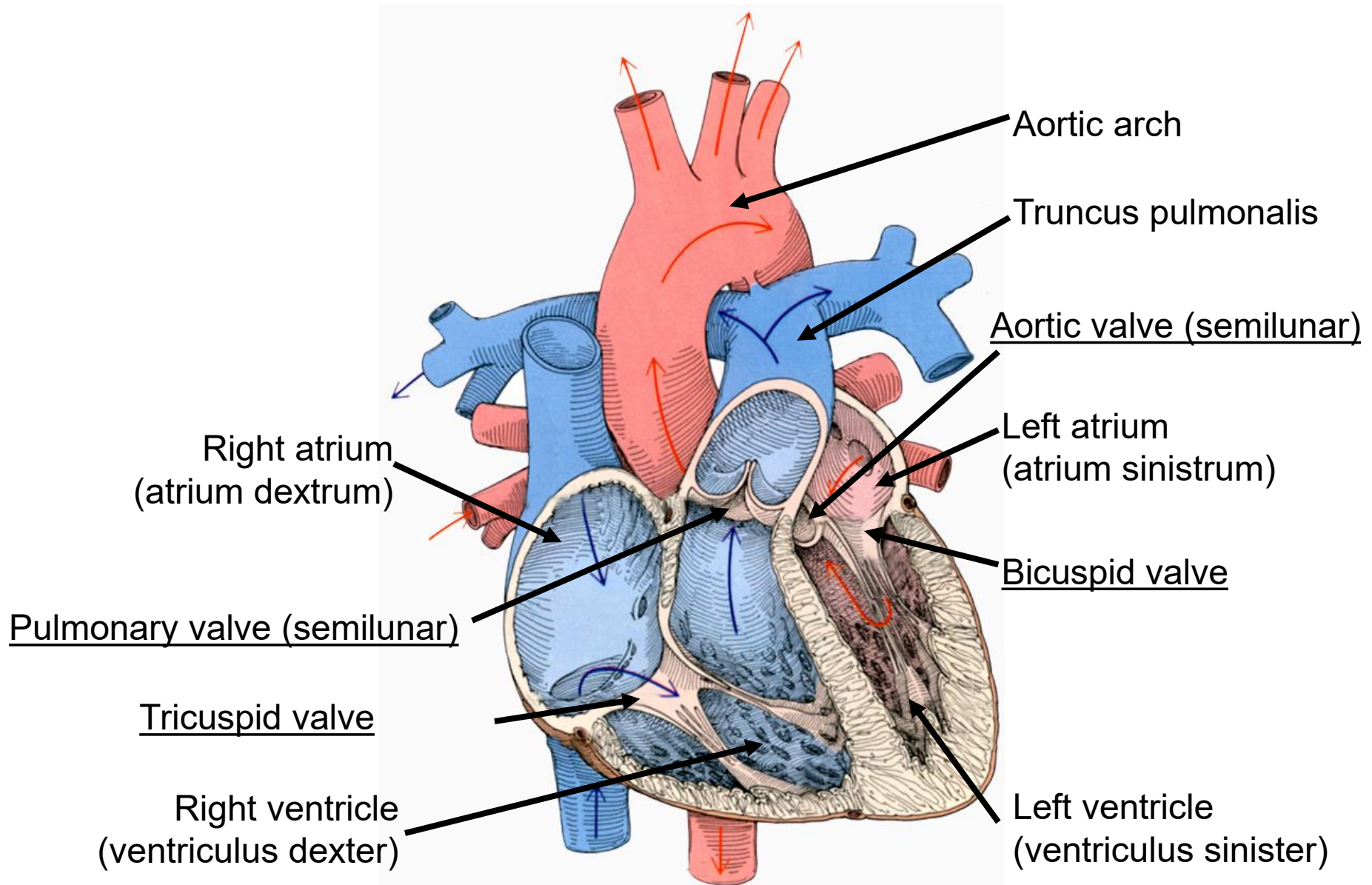


Heart: inferior and posterior surface



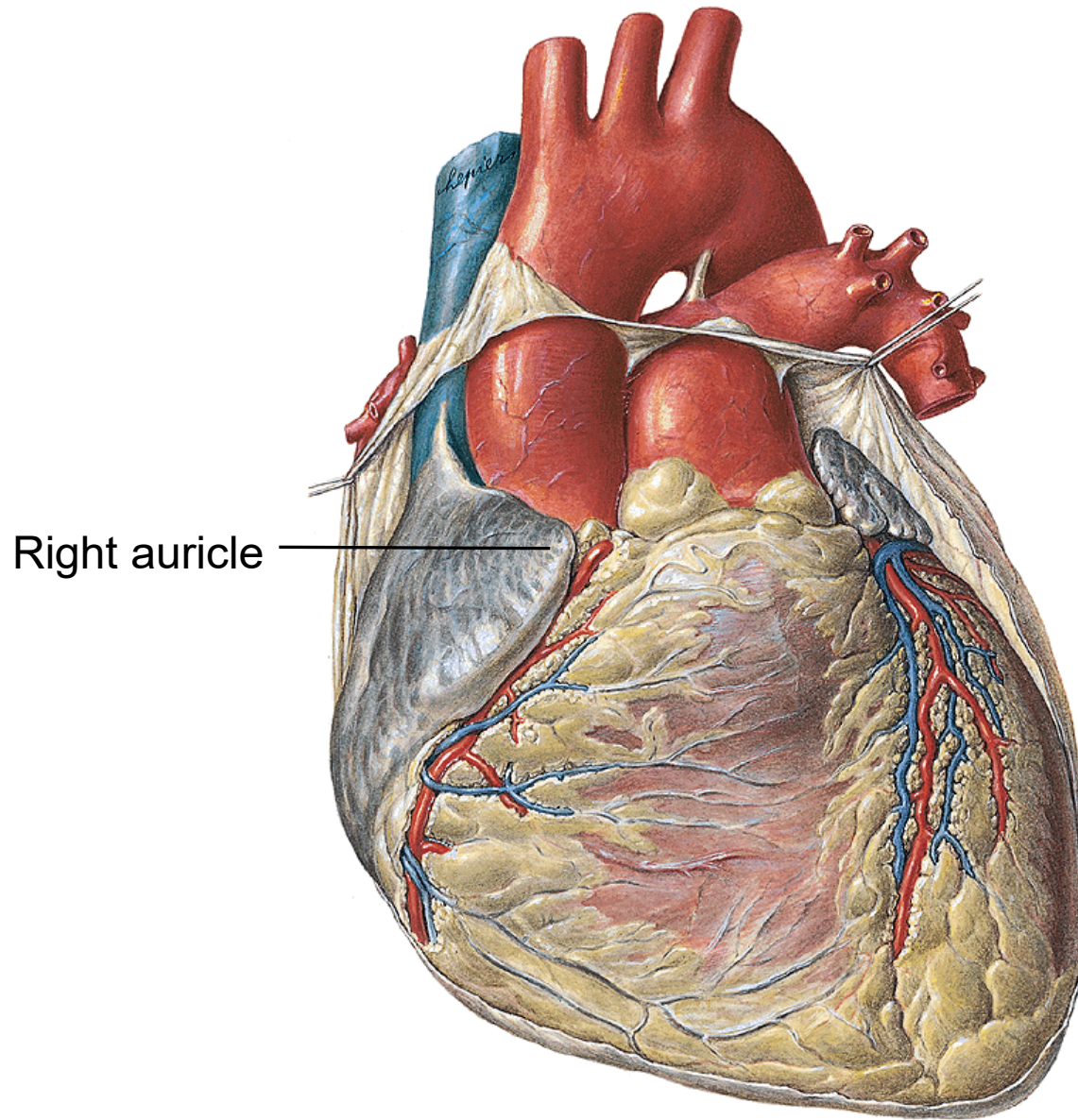
Cavities of the heart

Cavities of the heart



RIGHT ATRIUM

anterior view

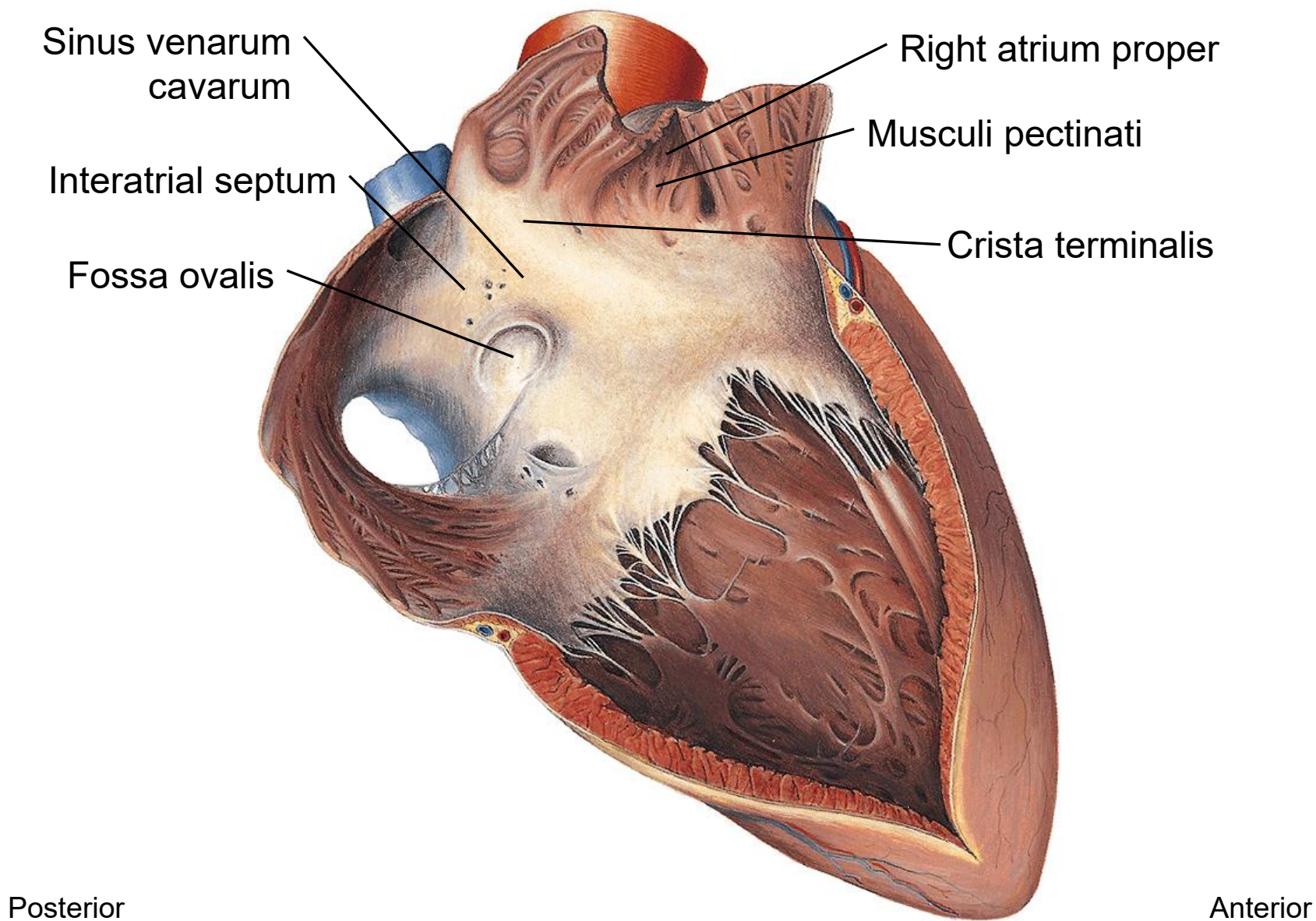


Right

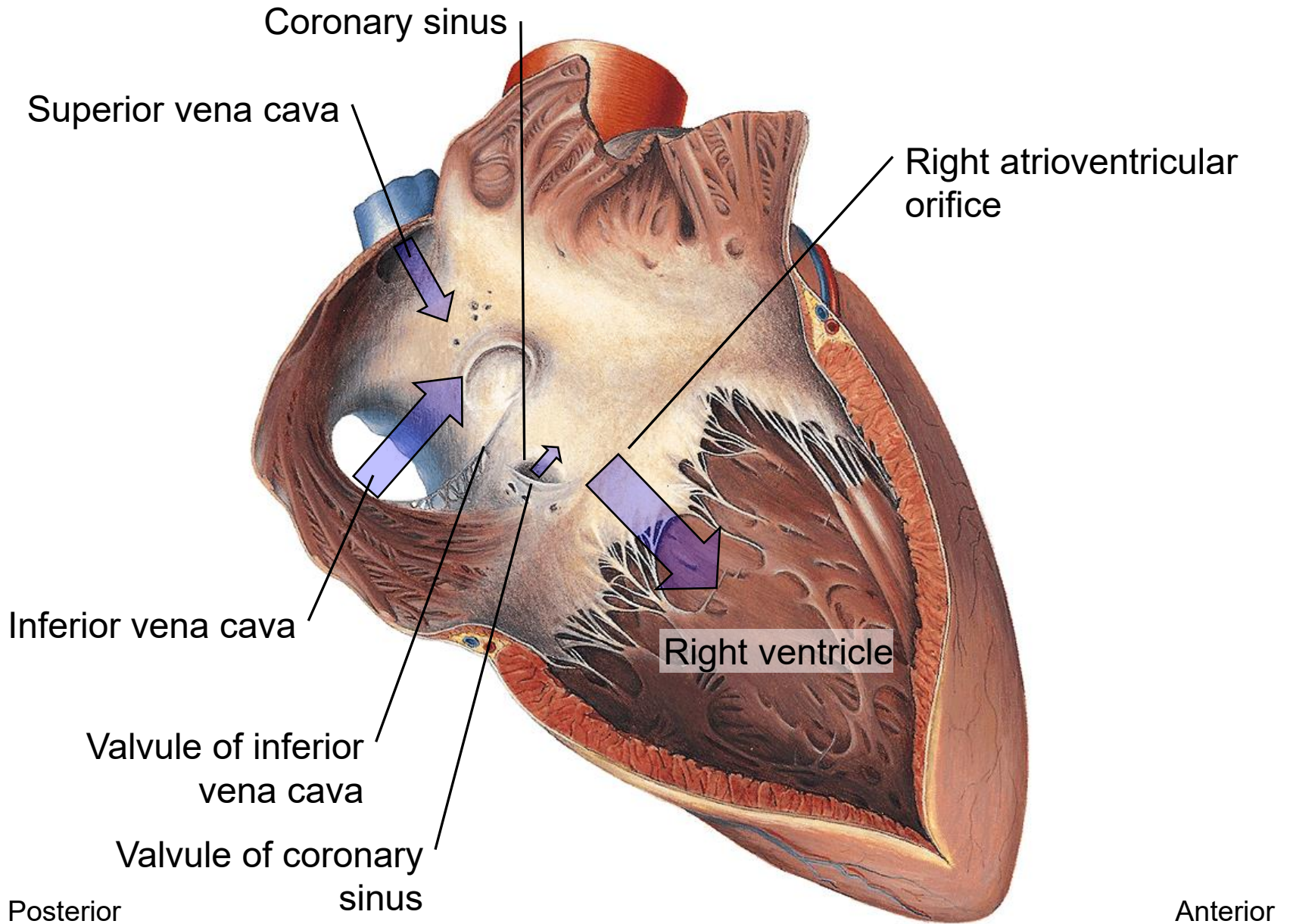
Left

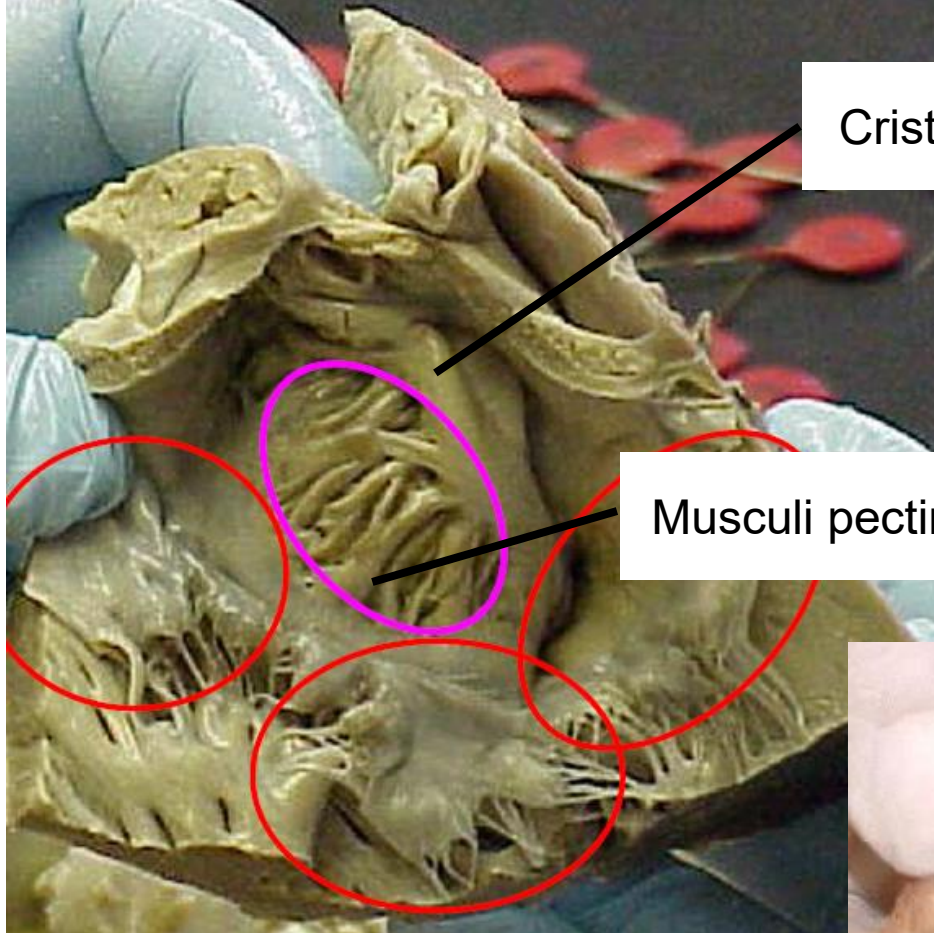
RIGHT ATRIUM

view from right



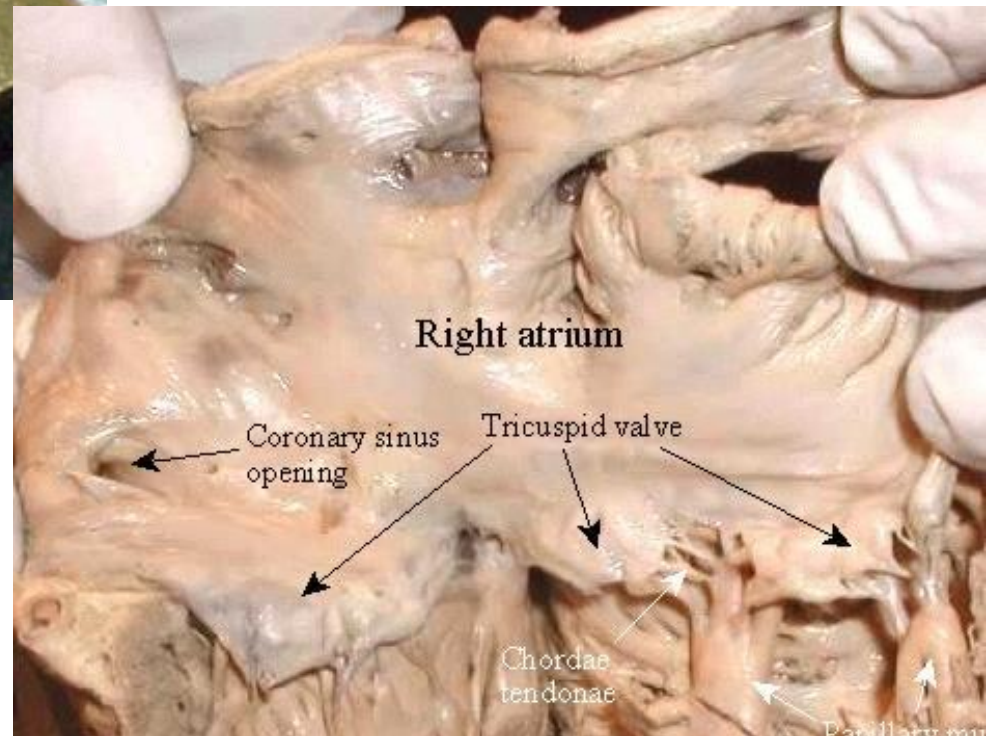
RIGHT ATRIUM view from right





Crista terminalis

Musculi pectinati



Right atrium

Coronary sinus
opening

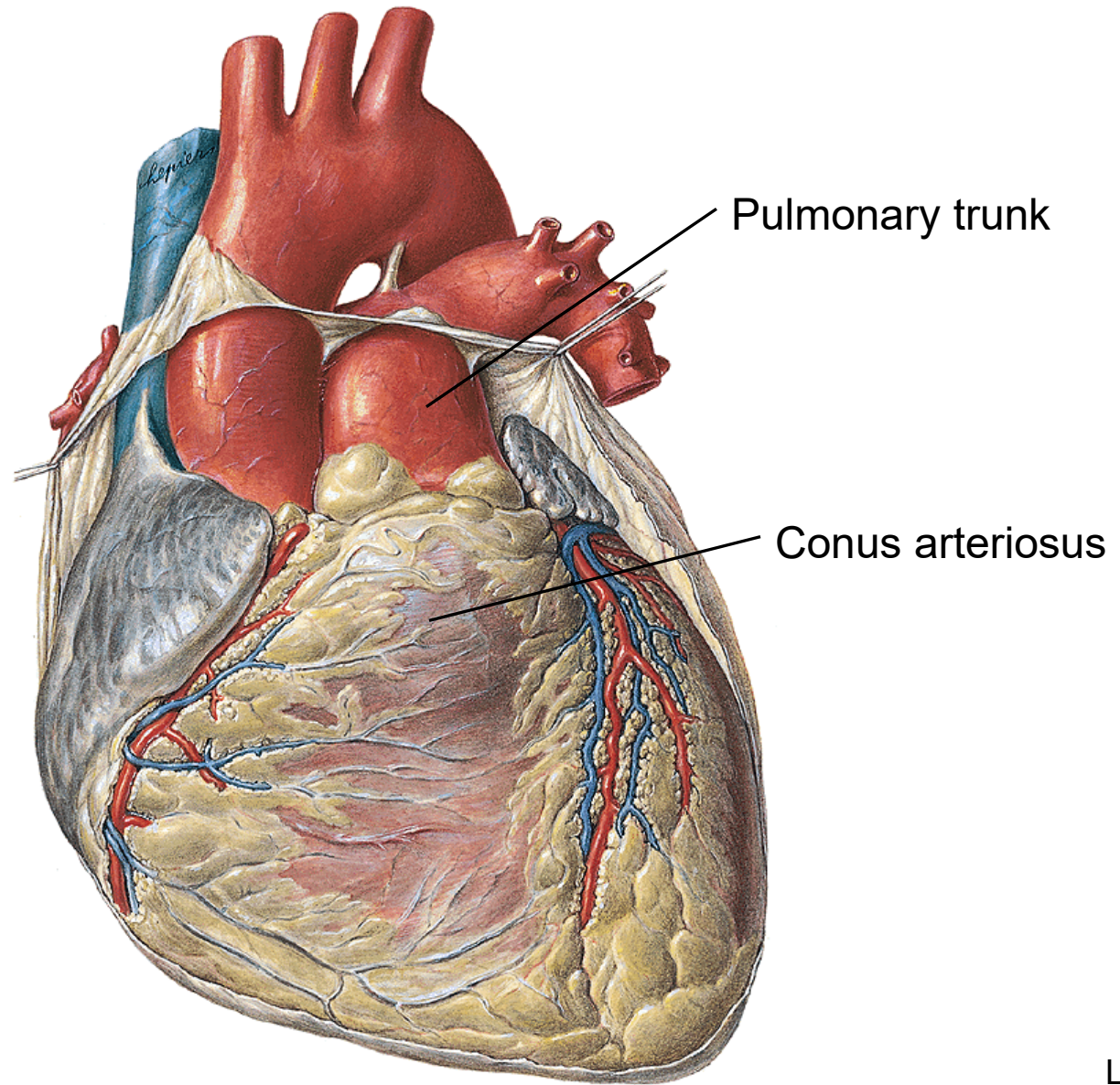
Tricuspid valve

Chordae
tendonae

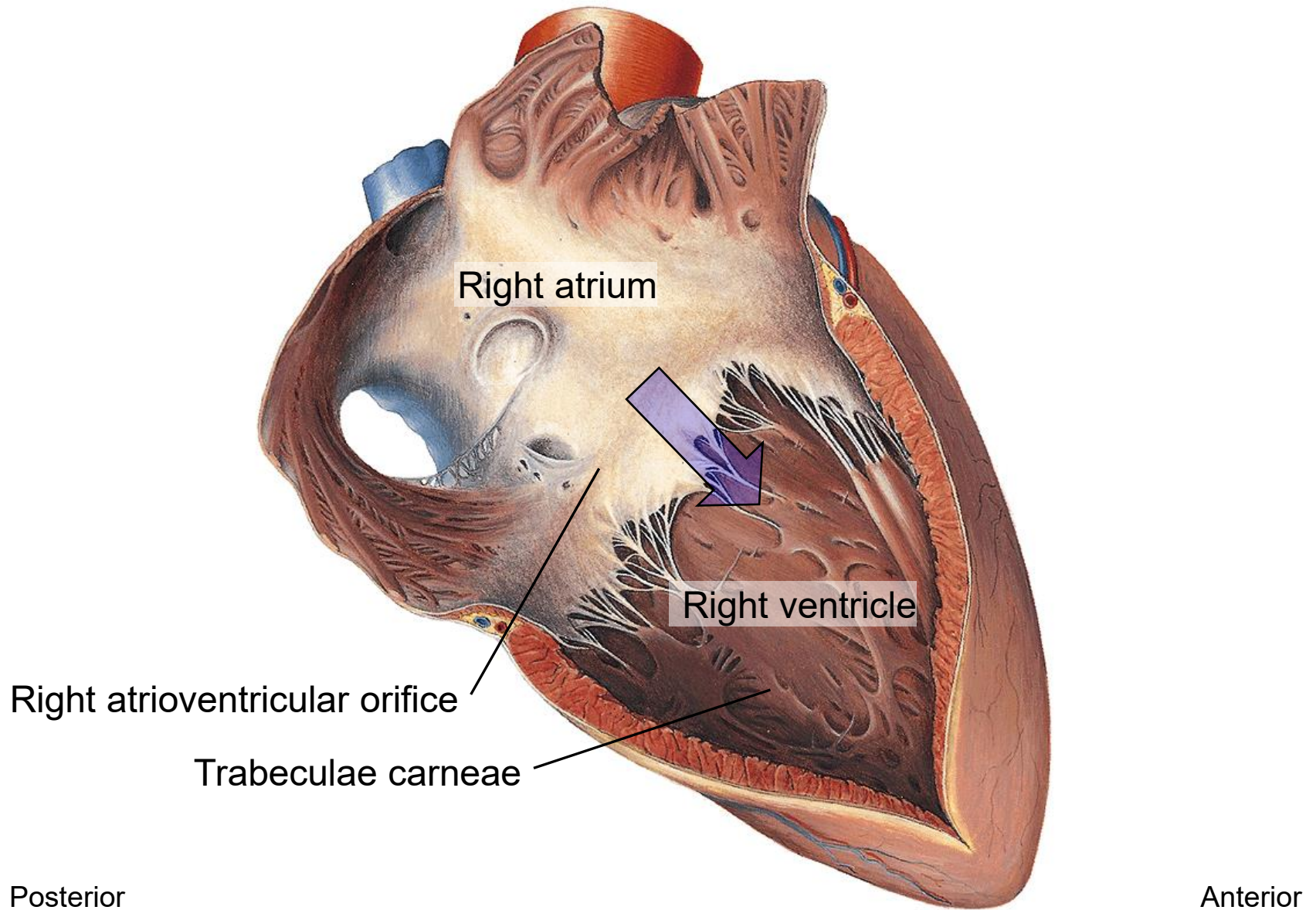
Papillary muscle

RIGHT VENTRICLE

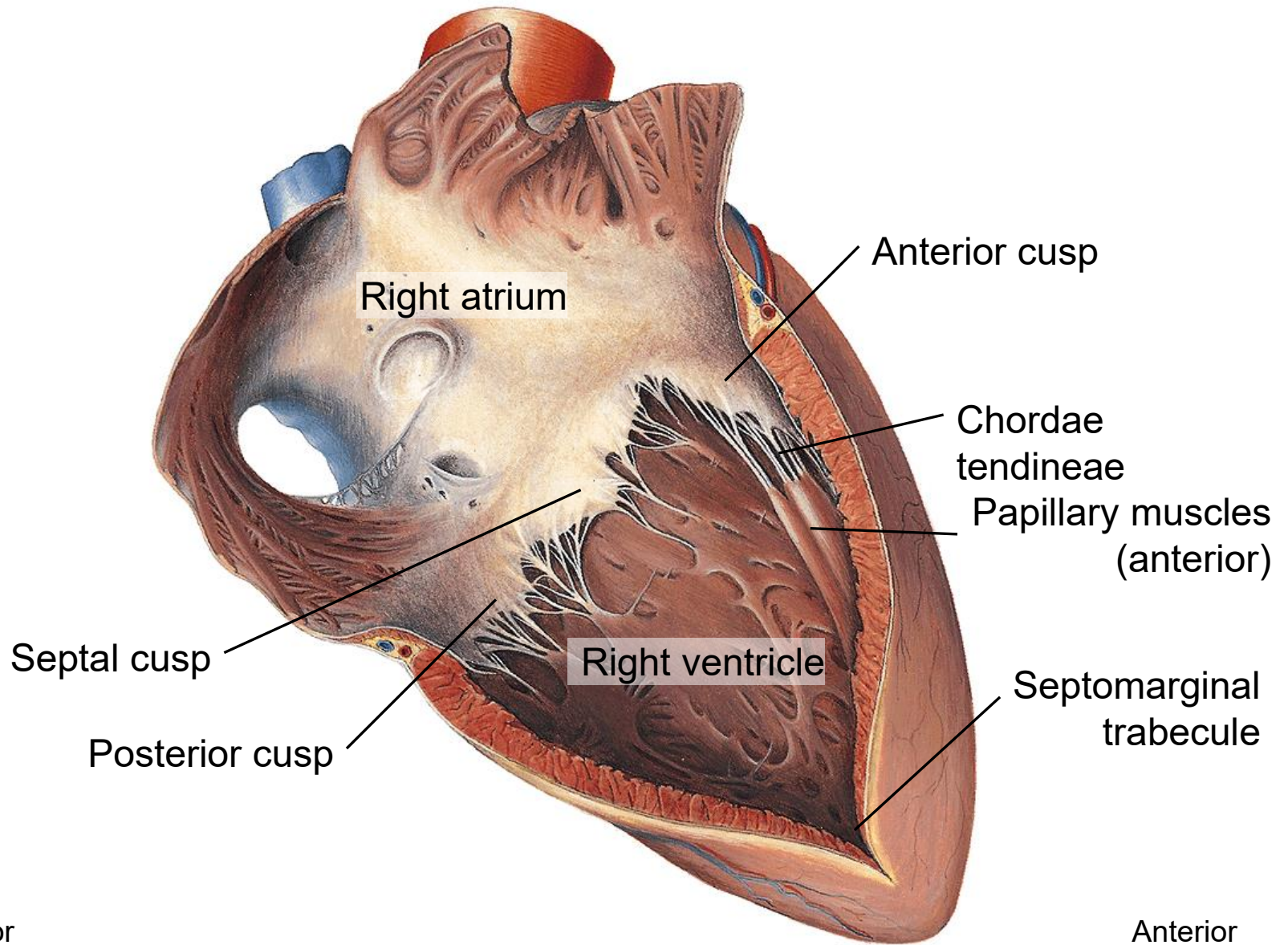
anterior view



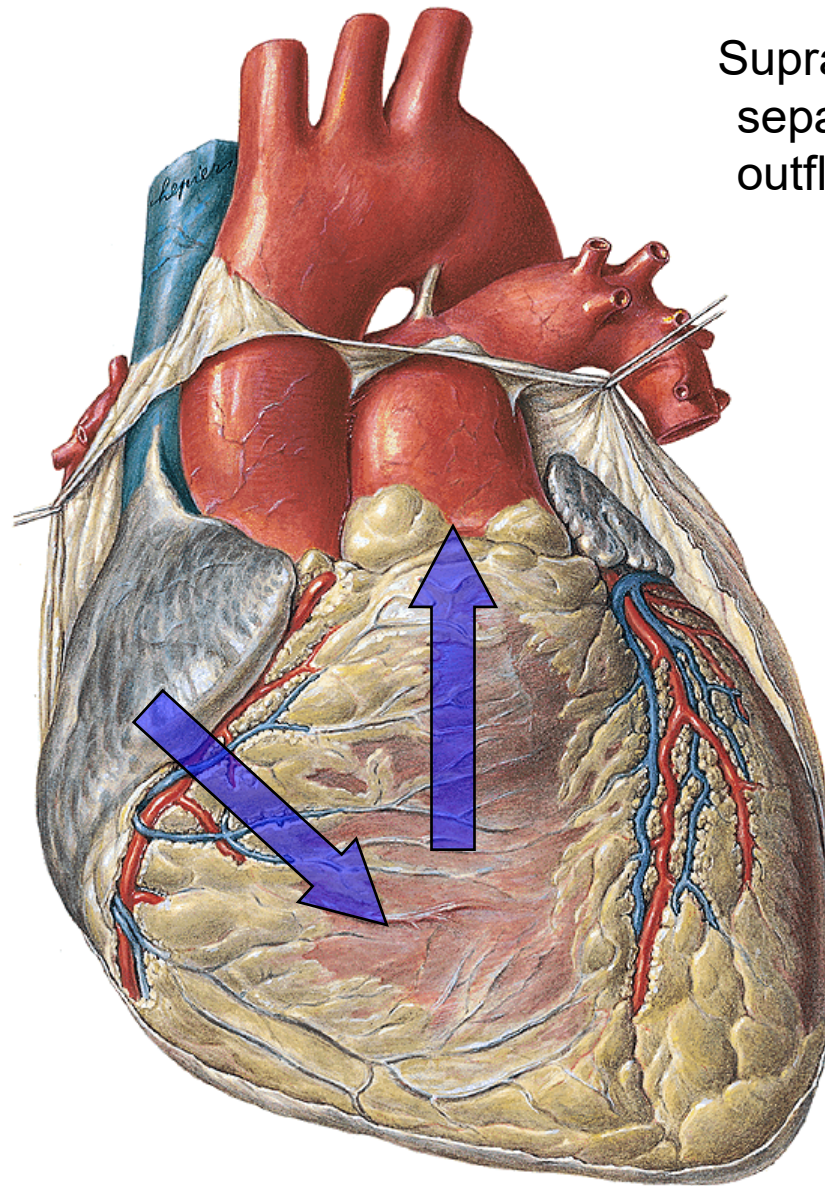
RIGHT VENTRICLE view from right



RIGHT VENTRICLE view from right



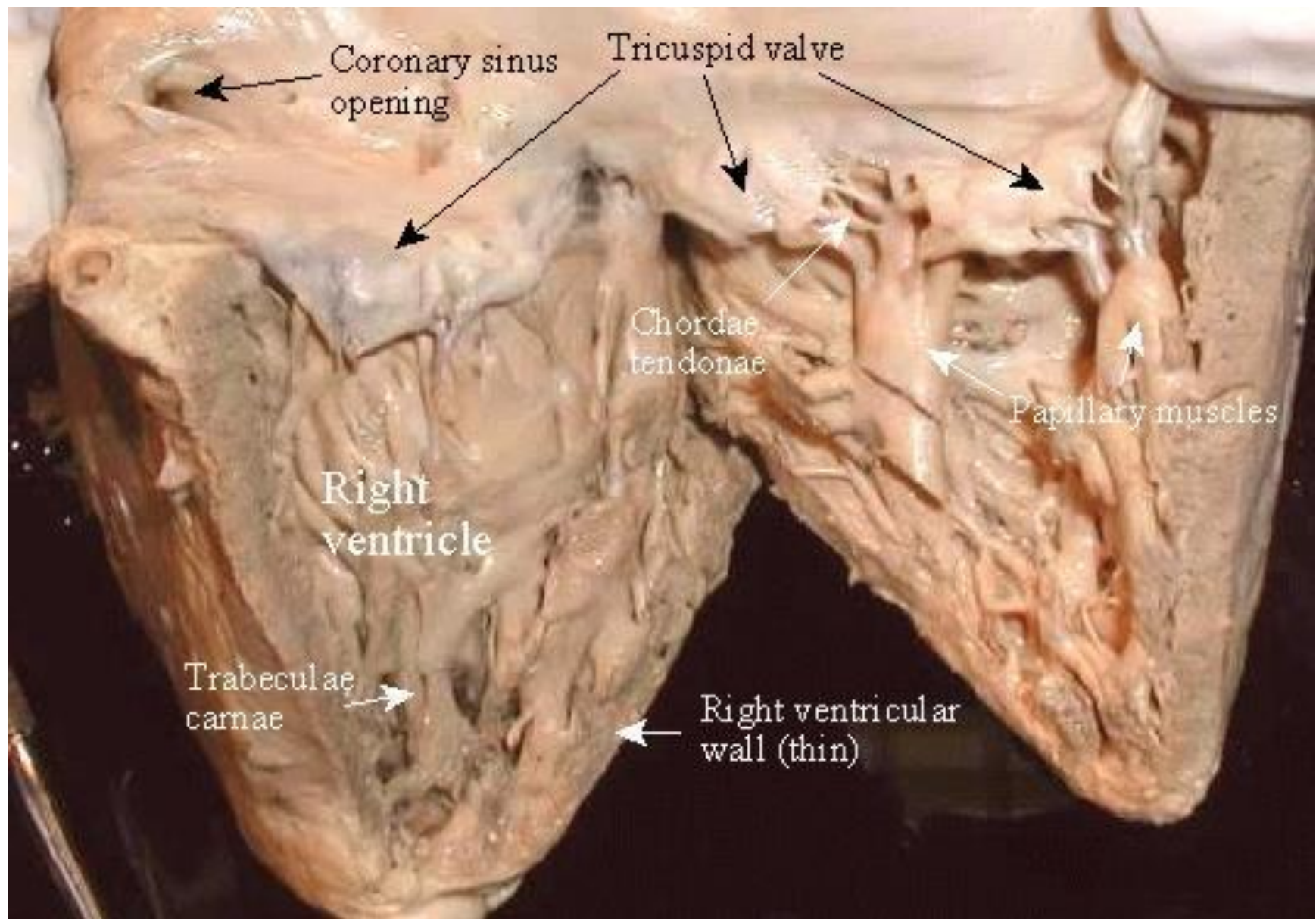
DIRECTION OF BLOOD FLOW



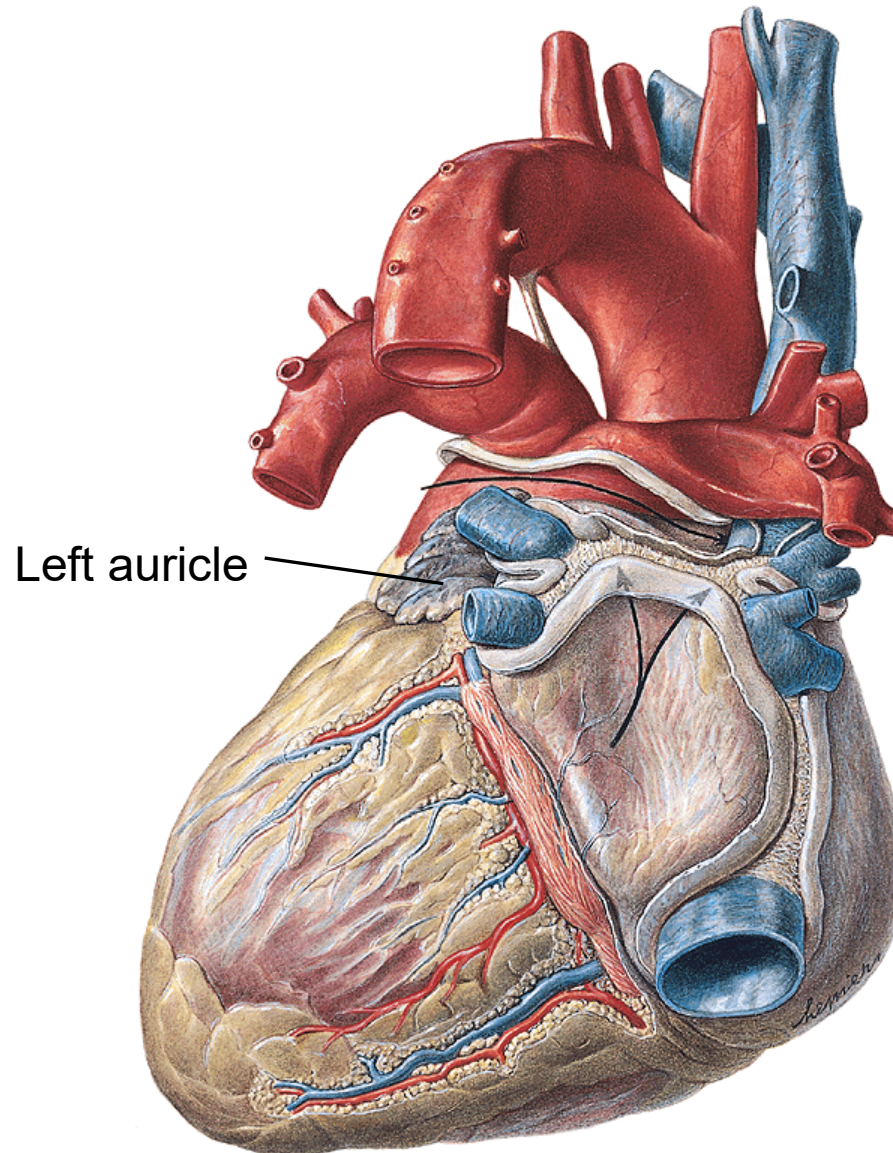
Supraventricular crest:
separates inflow and
outflow tracts

Right

Left



LEFT ATRIUM posterior view

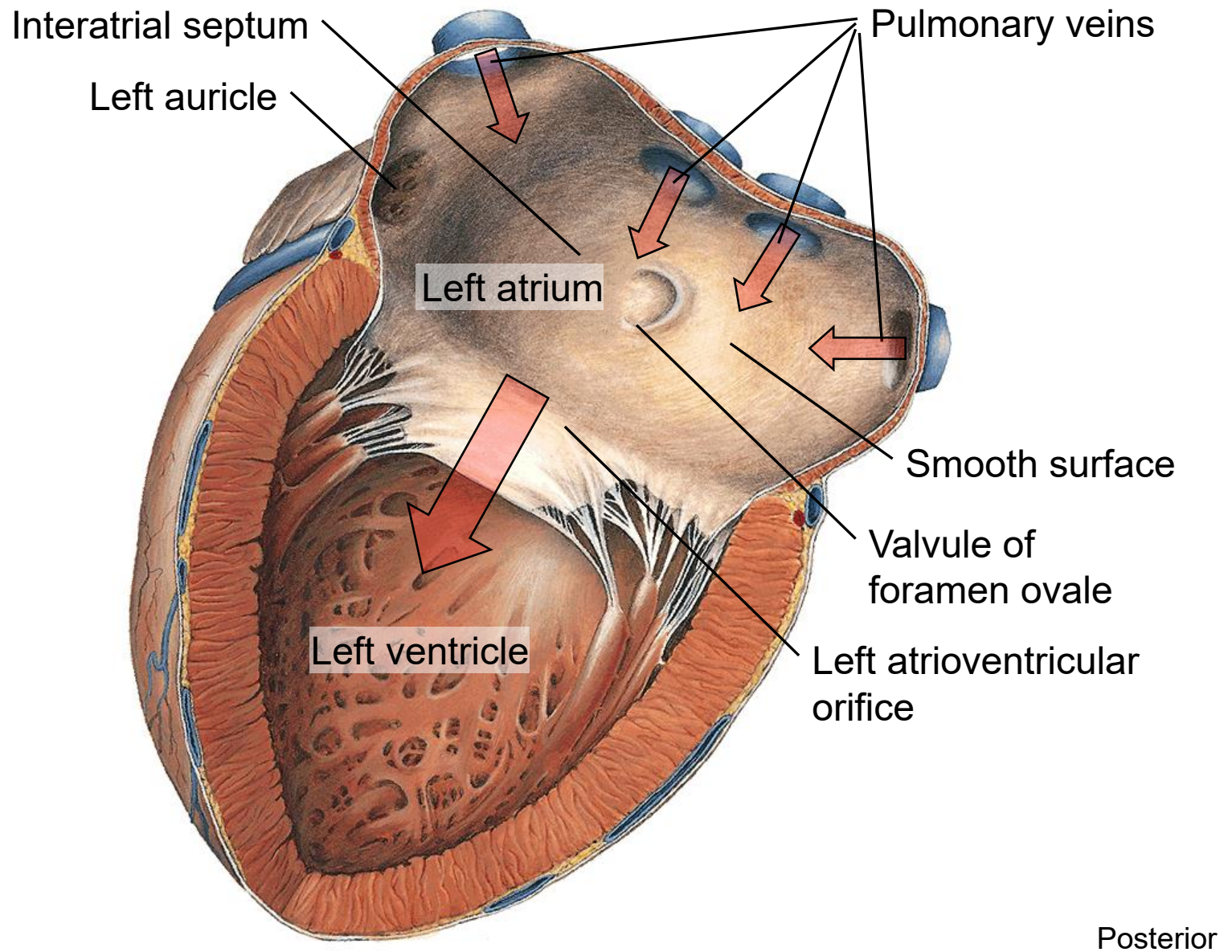


Left

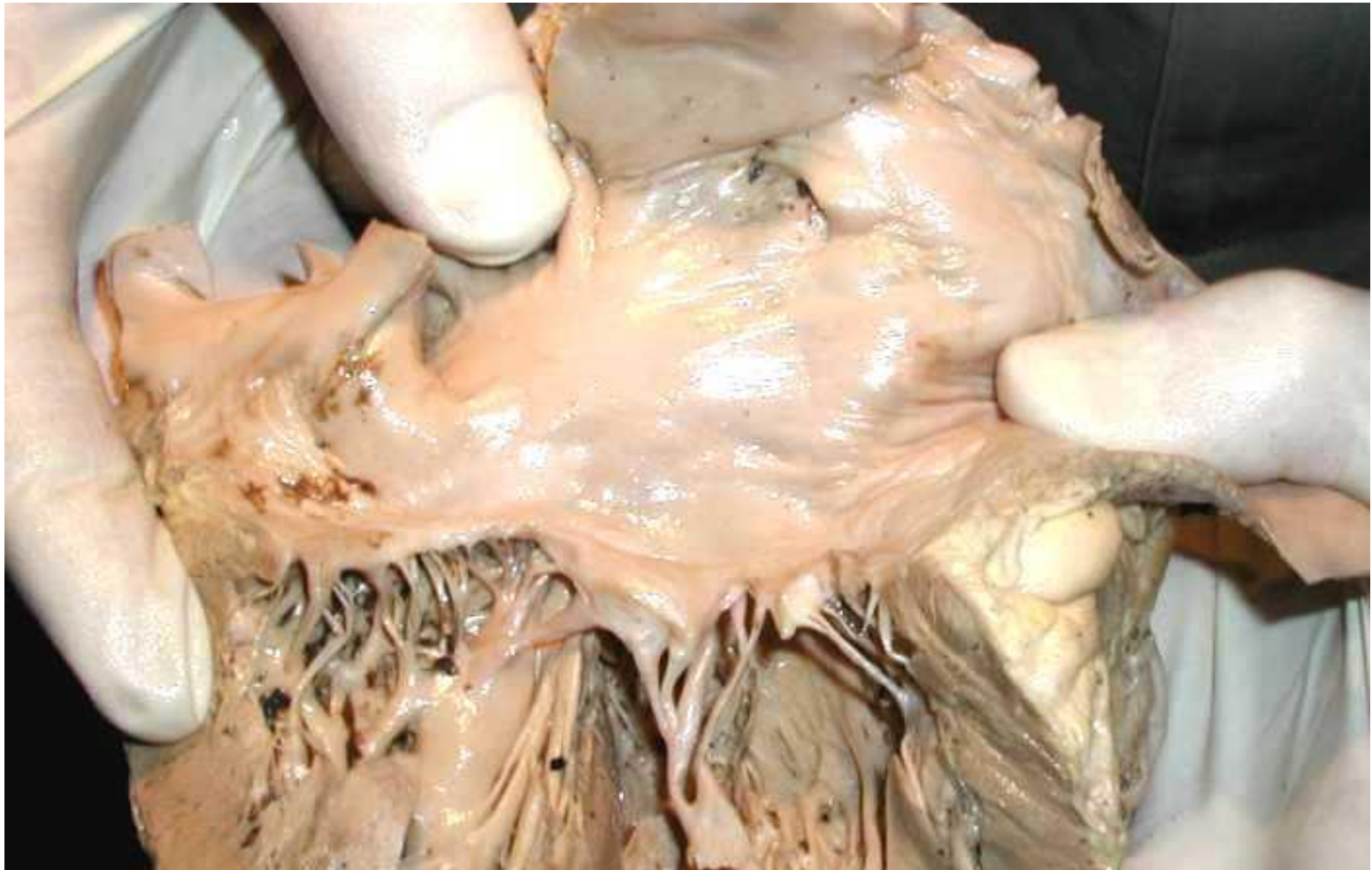
Right

LEFT ATRIUM

view from left

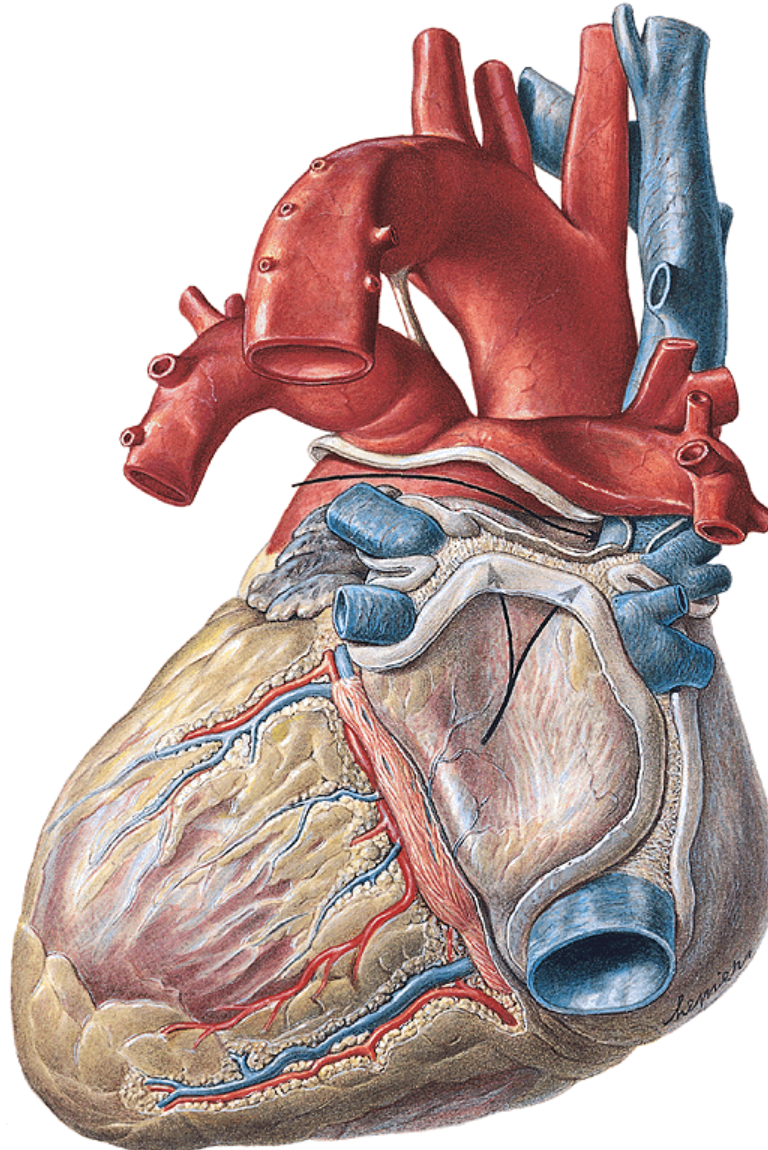


LEFT ATRIUM view from left



LEFT VENTRICLE

posterior view

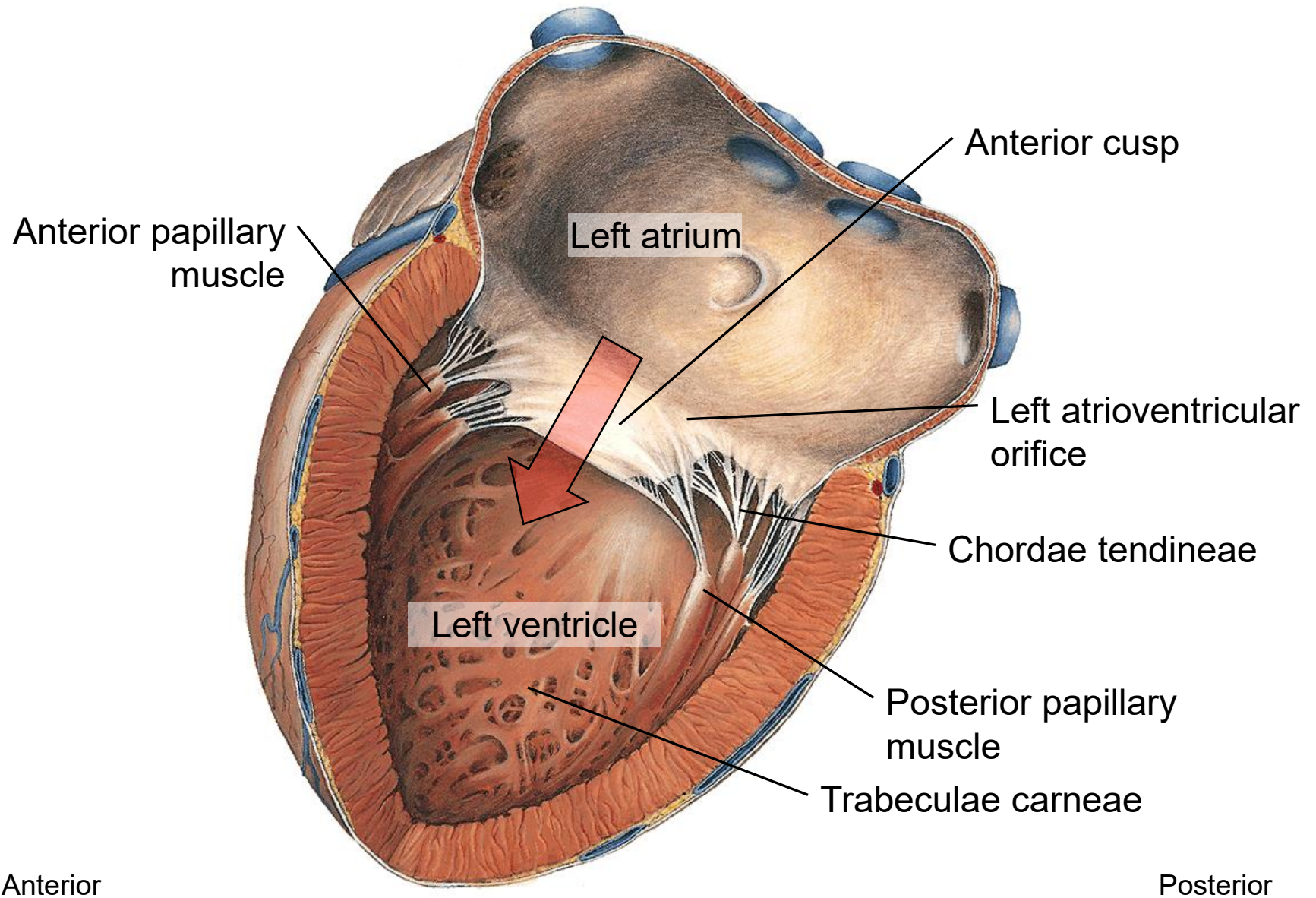


Left

Right

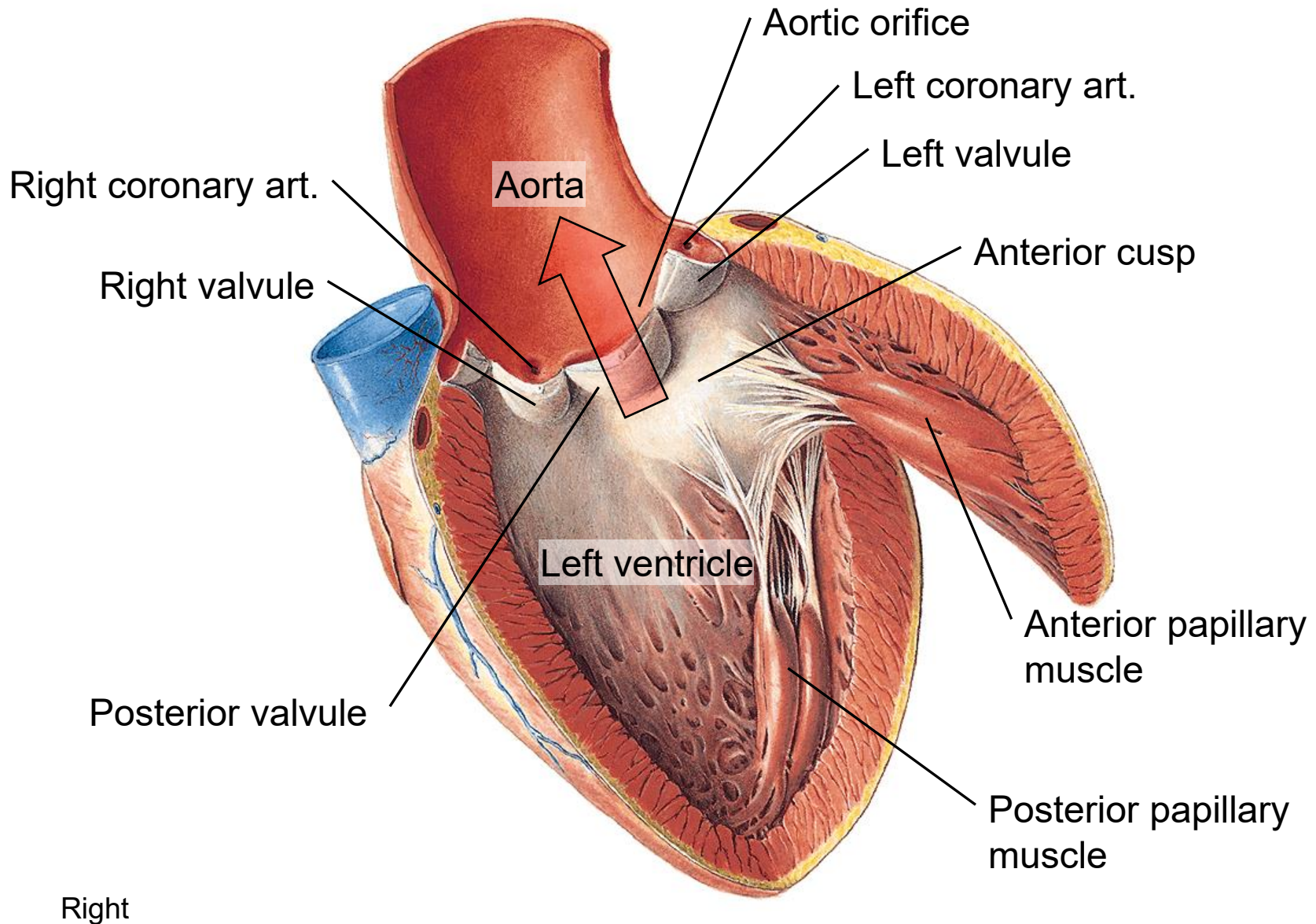
LEFT VENTRICLE

view from left

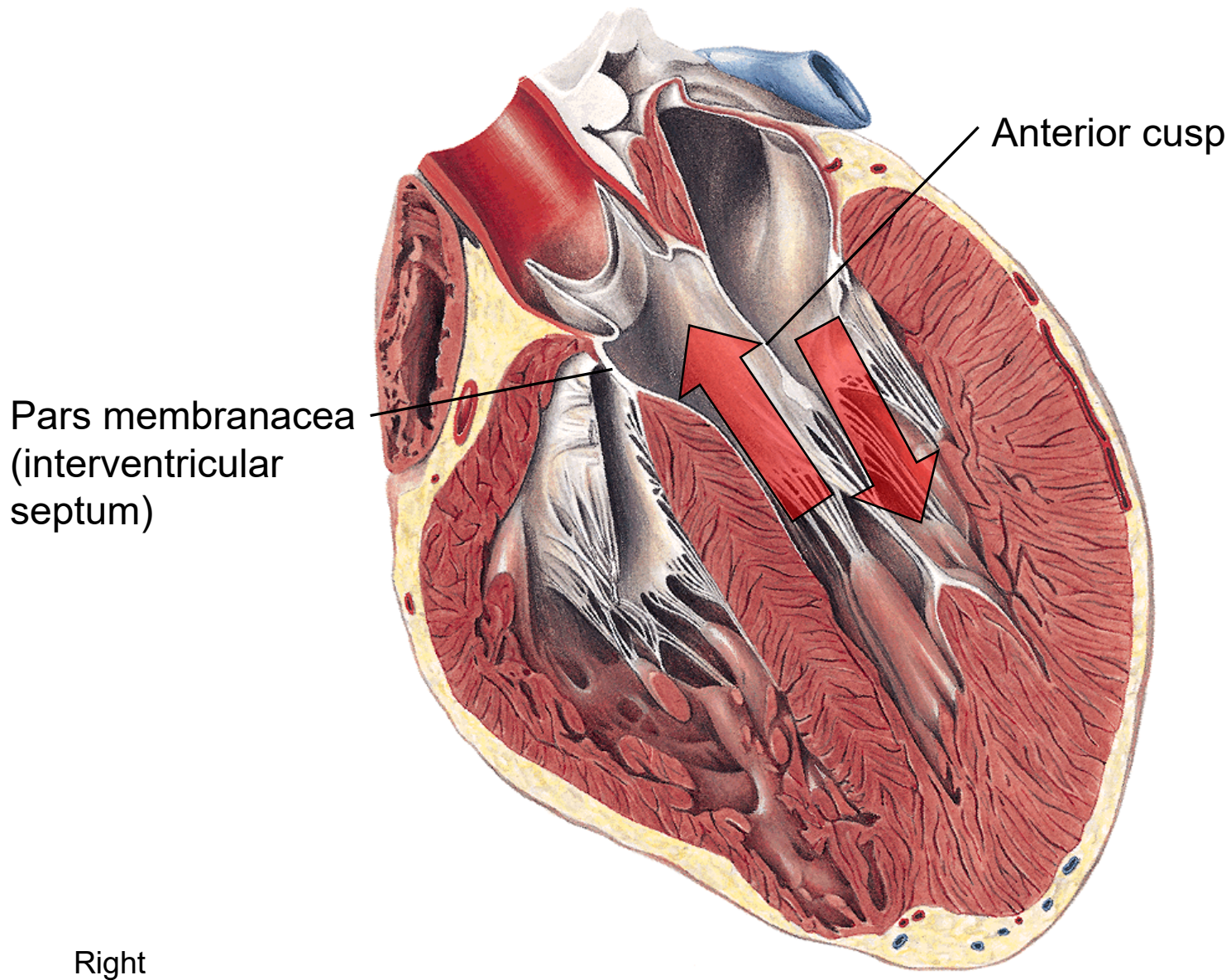


LEFT VENTRICLE

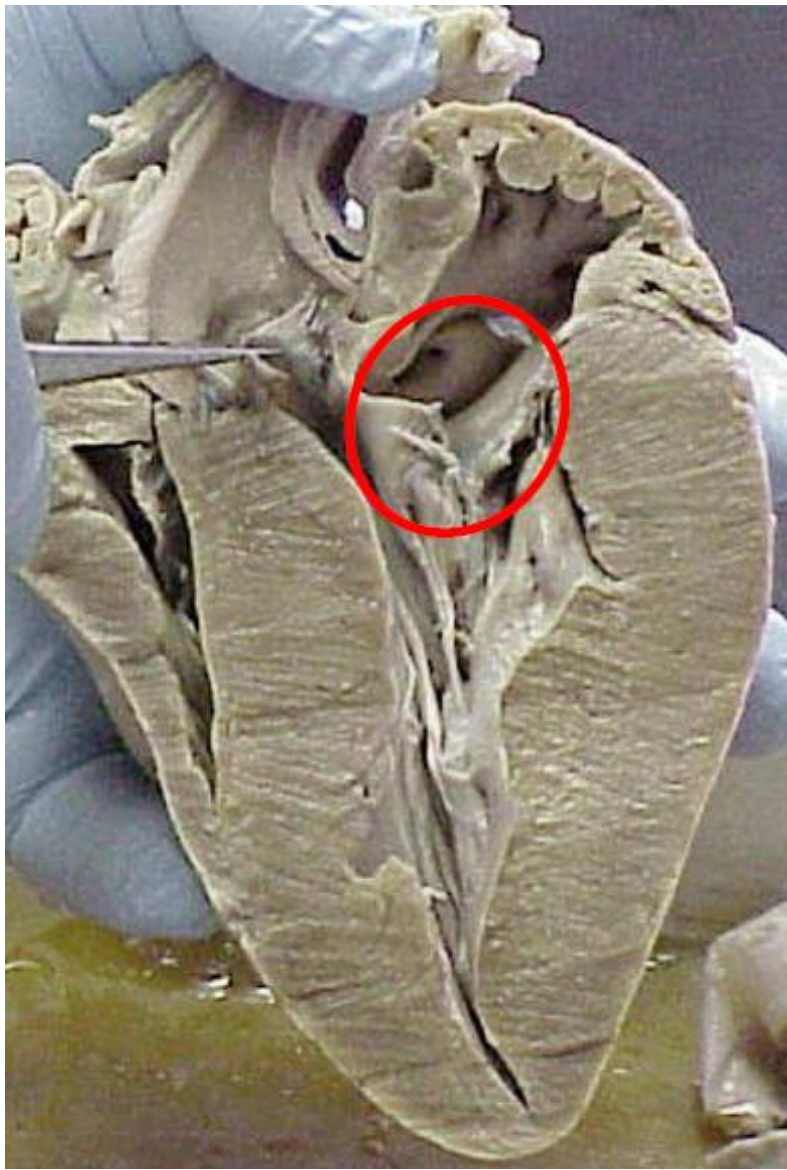
anterior view

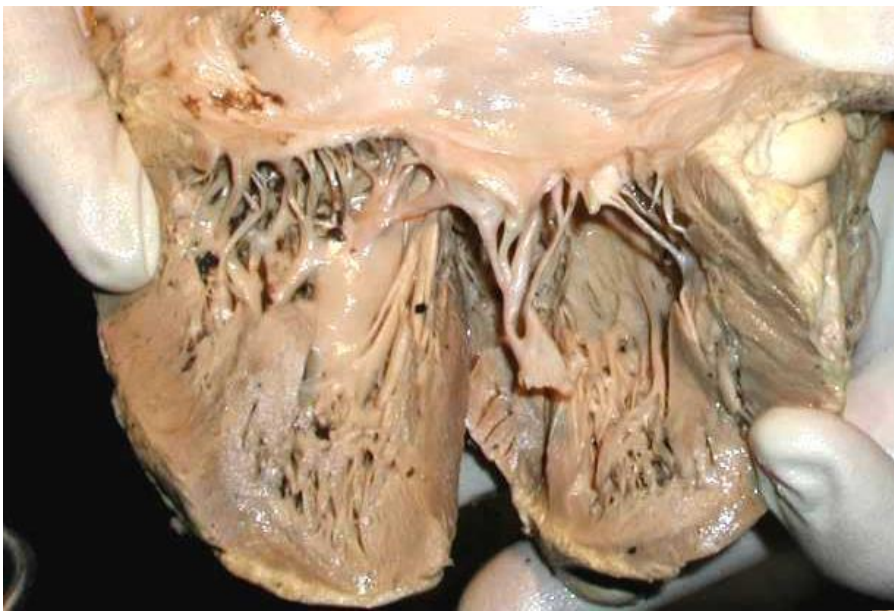


DIRECTION OF BLOOD FLOW

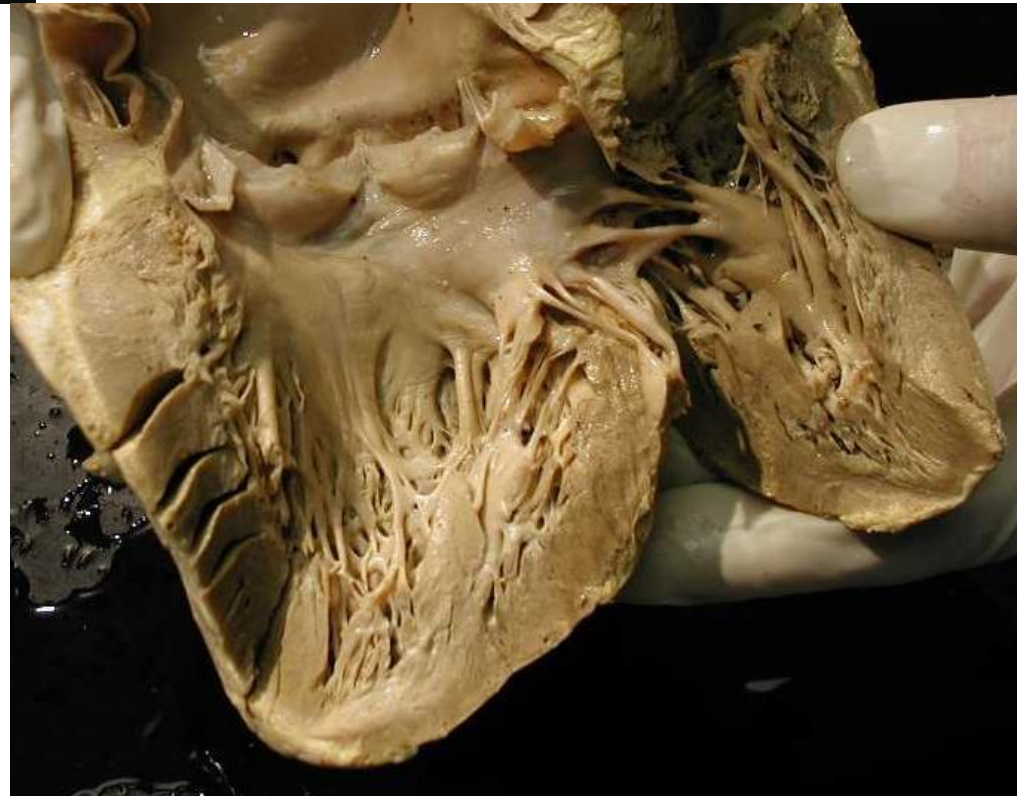


**LEFT VENTRICLE, LEFT ATRIUM,
MITRAL VALVE
view from left**





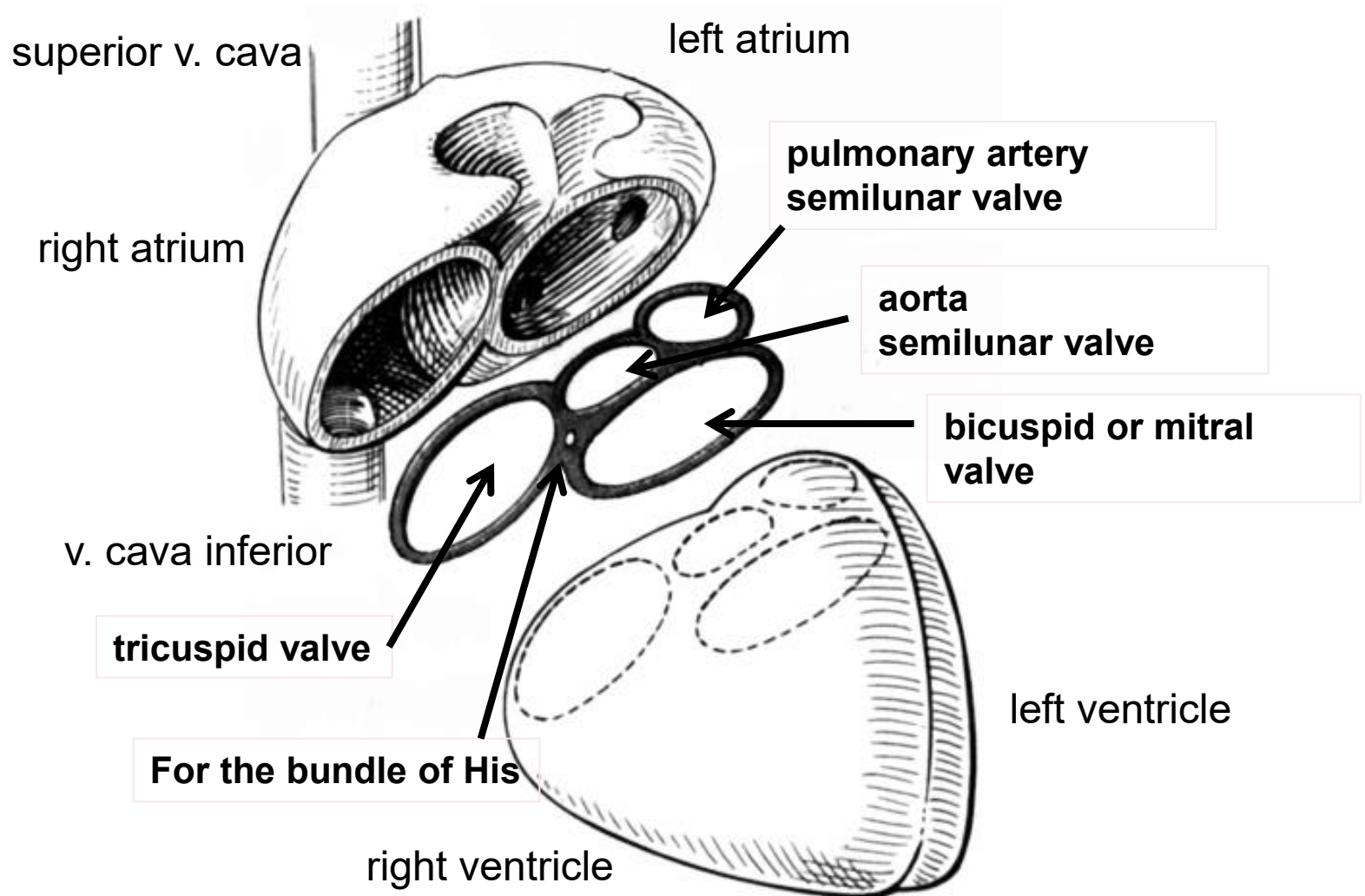
**LEFT VENTRICLE,
MITRAL VALVE
AORTA VALVE
view from left**



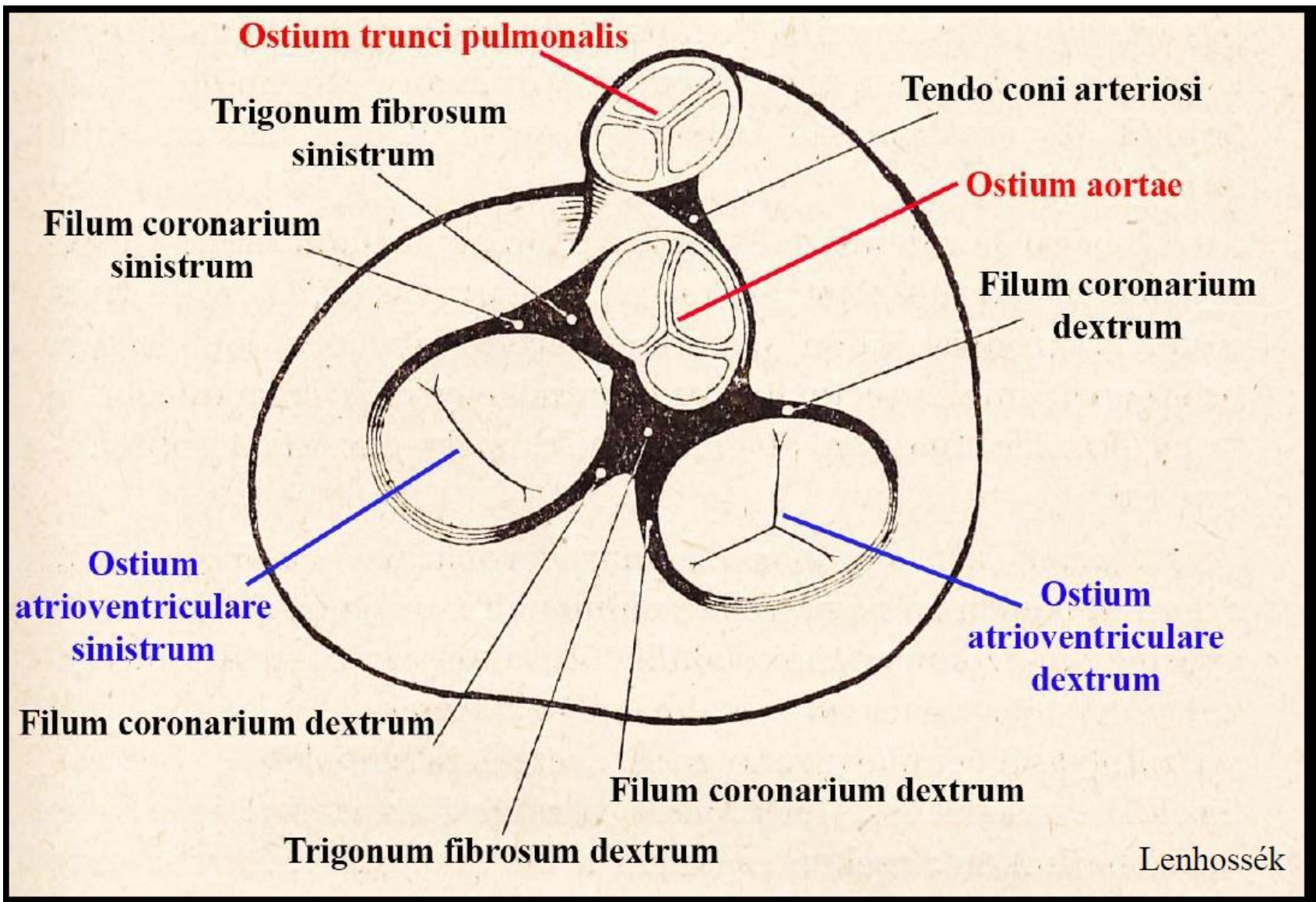
Cardiac skeleton
Anulus fibrosus

Cardiac skeleton

The four valves of the heart are anchored in dense fibrous connective tissue between them, they form a unit, the cardiac skeleton to which the atria and ventricles are attached.



Anulus fibrosus

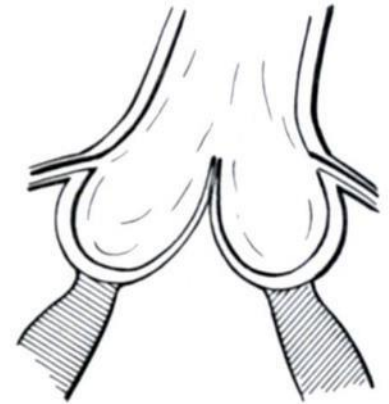


Valves of the heart



Cardiac valves

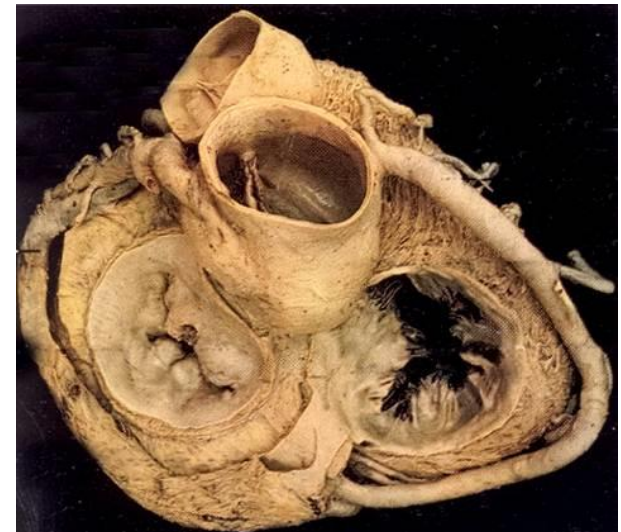
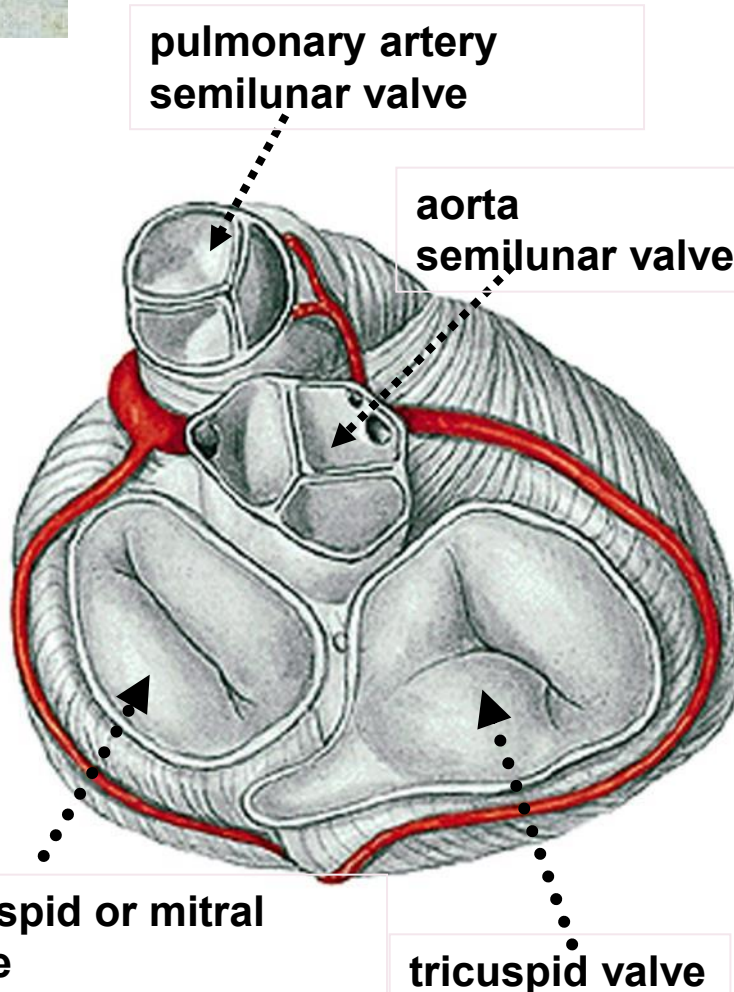
arise from the anuli fibrosi



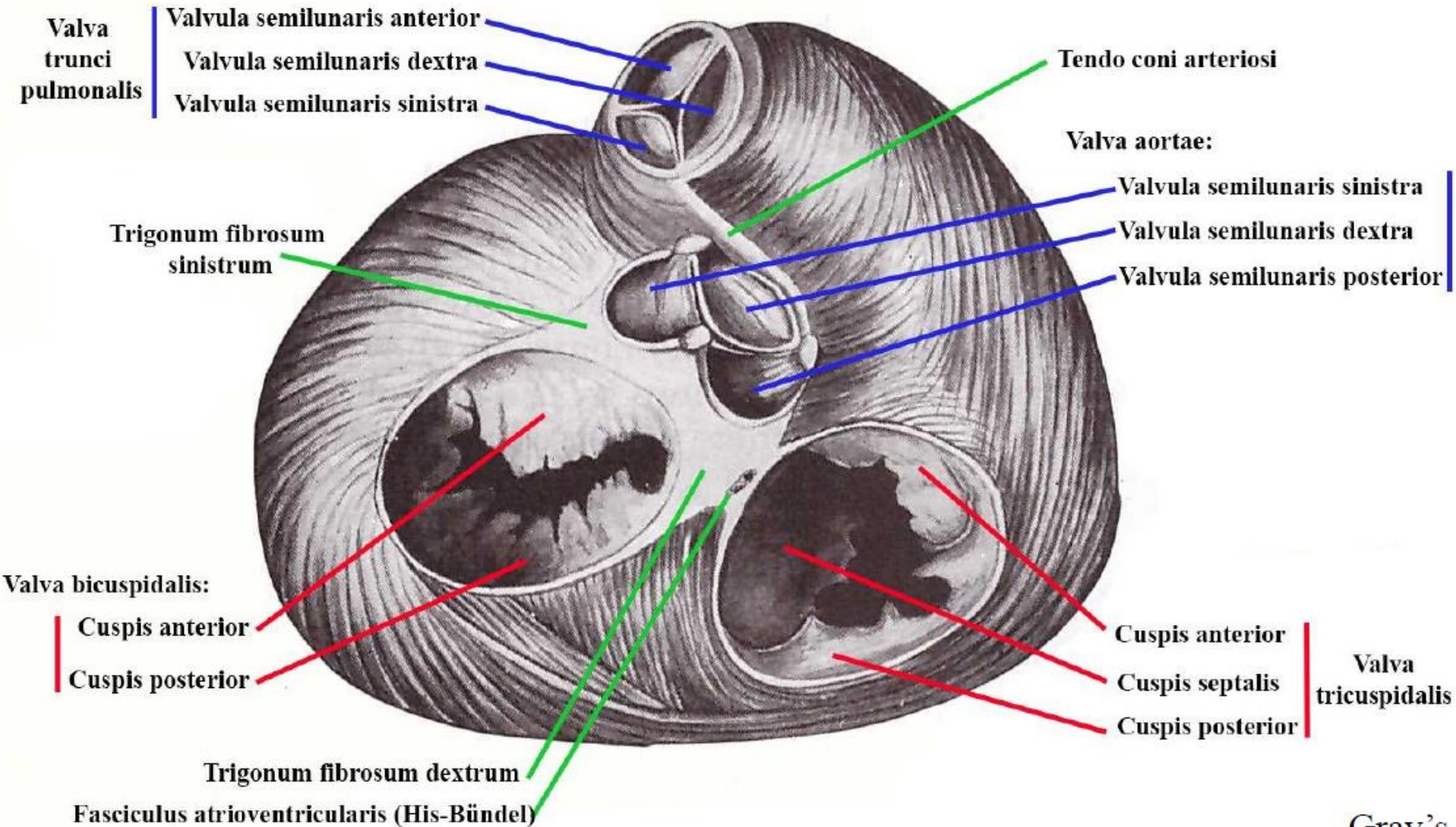
semilunar valve

- Aorta
- Truncus pulmonalis

cuspid valve
left and right
atrioventricular
openings

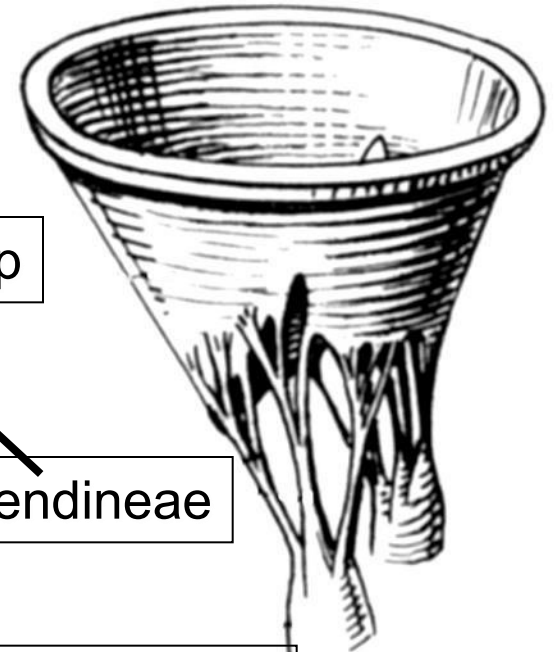
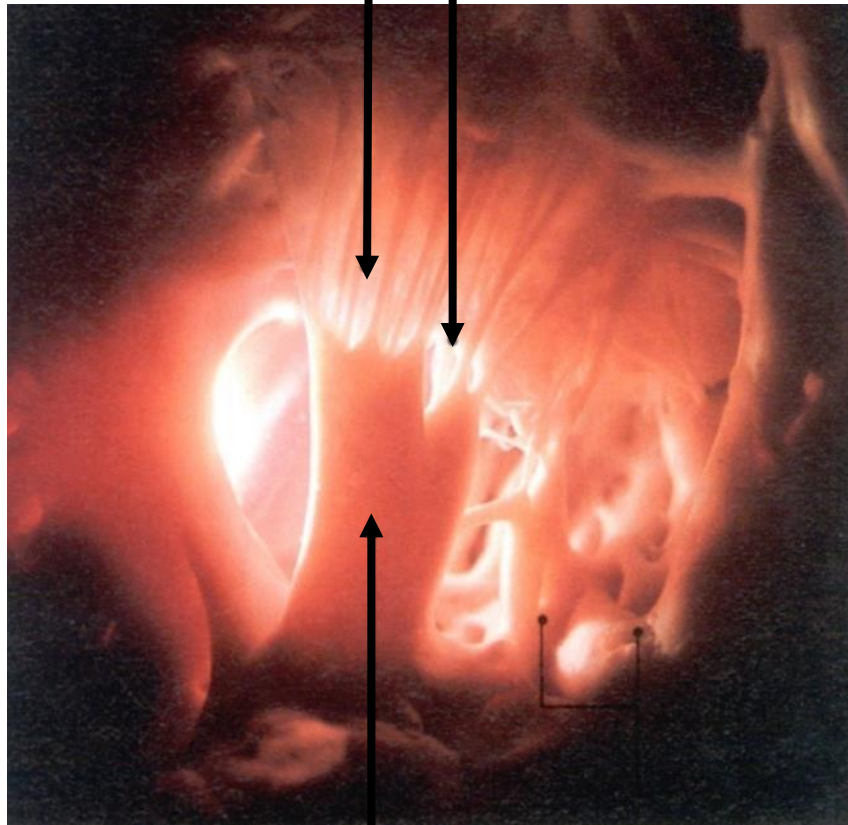


CARDIAC VALVES



VALVES OF HEART

cuspid valves



cusps

chordae tendineae

papillary muscle

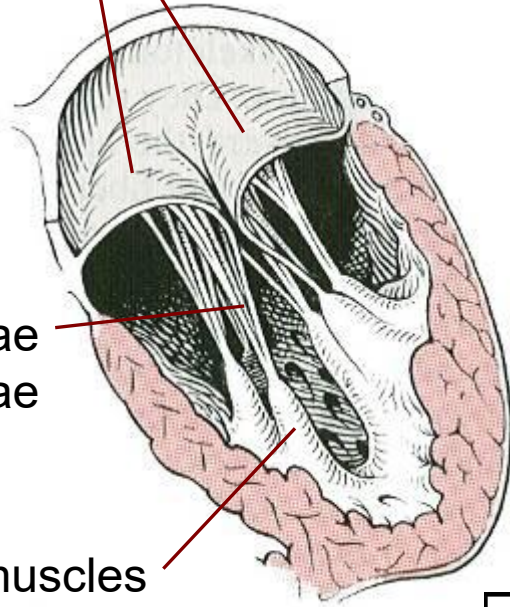
VALVES OF HEART

cuspid valves

Cusps
(2 or 3)

Chordae
tendineae

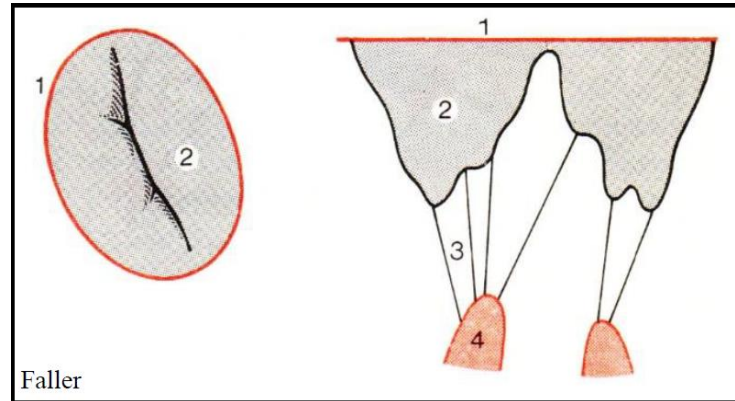
Papillary muscles



Closed



Open

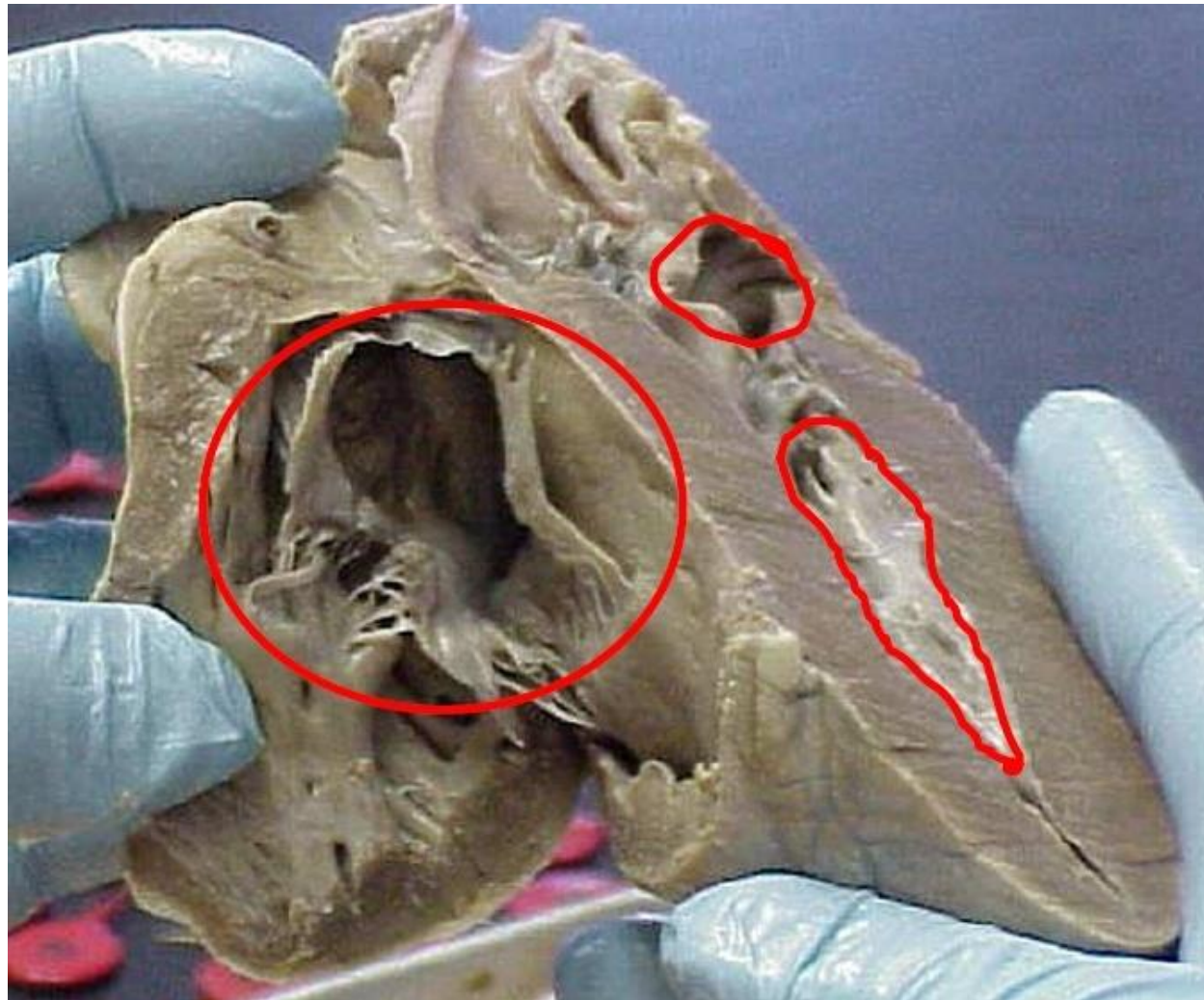


Tricuspid valve

Bicuspid valve (mitral)

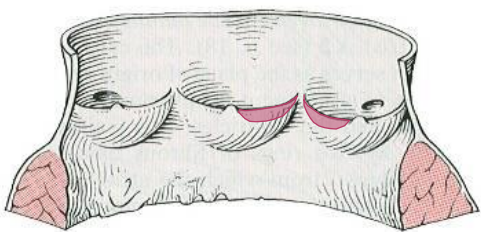
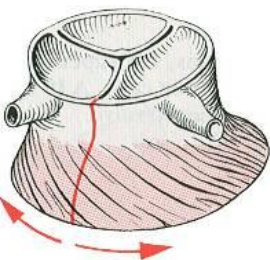
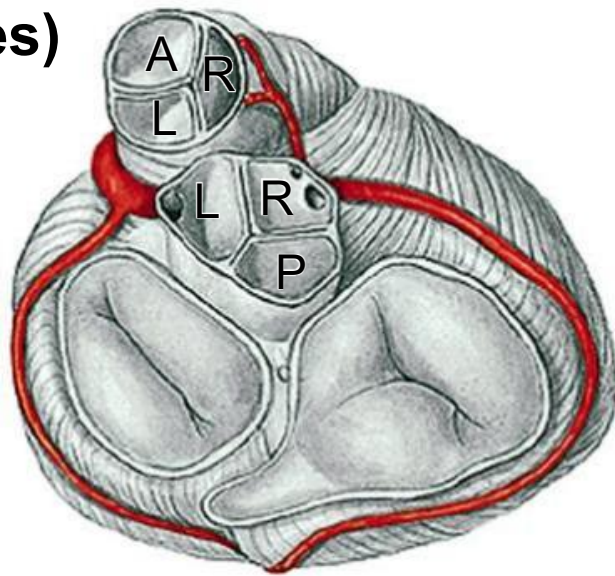


Tricuspid valve



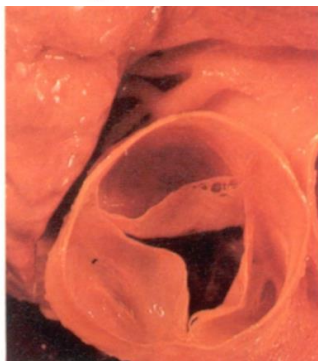
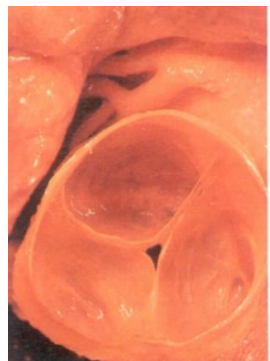
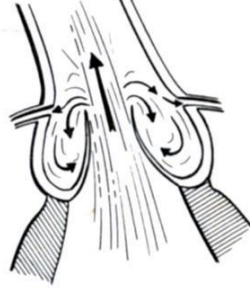
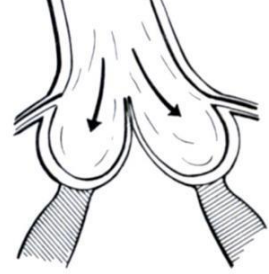
CARDIAC VALVES (semilunar valves)

aortic opening and
pulmonary opening

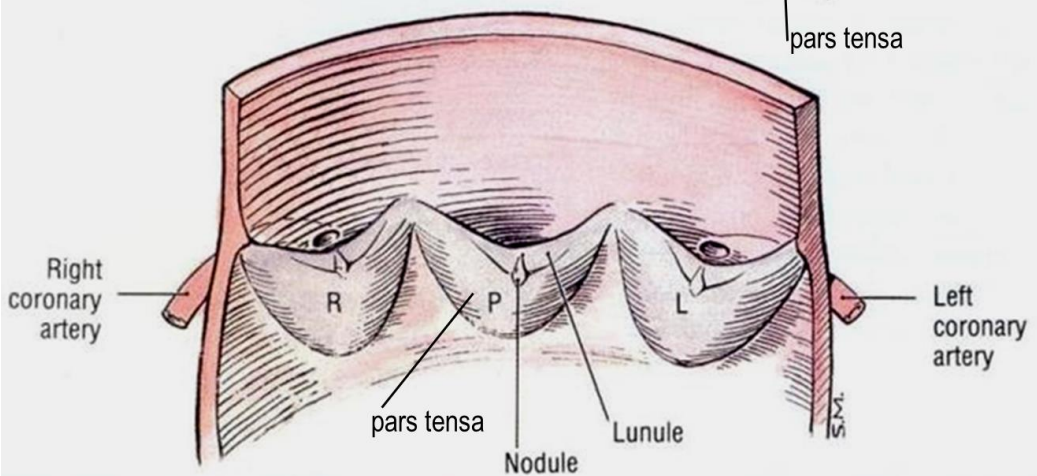
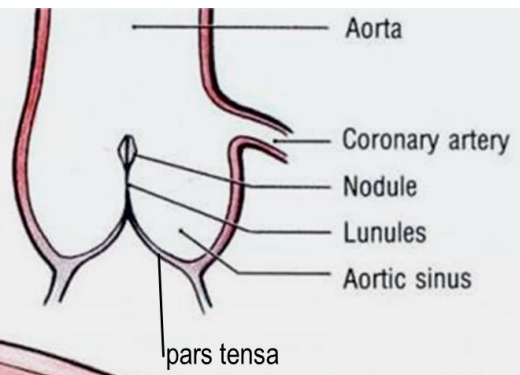


closed

open

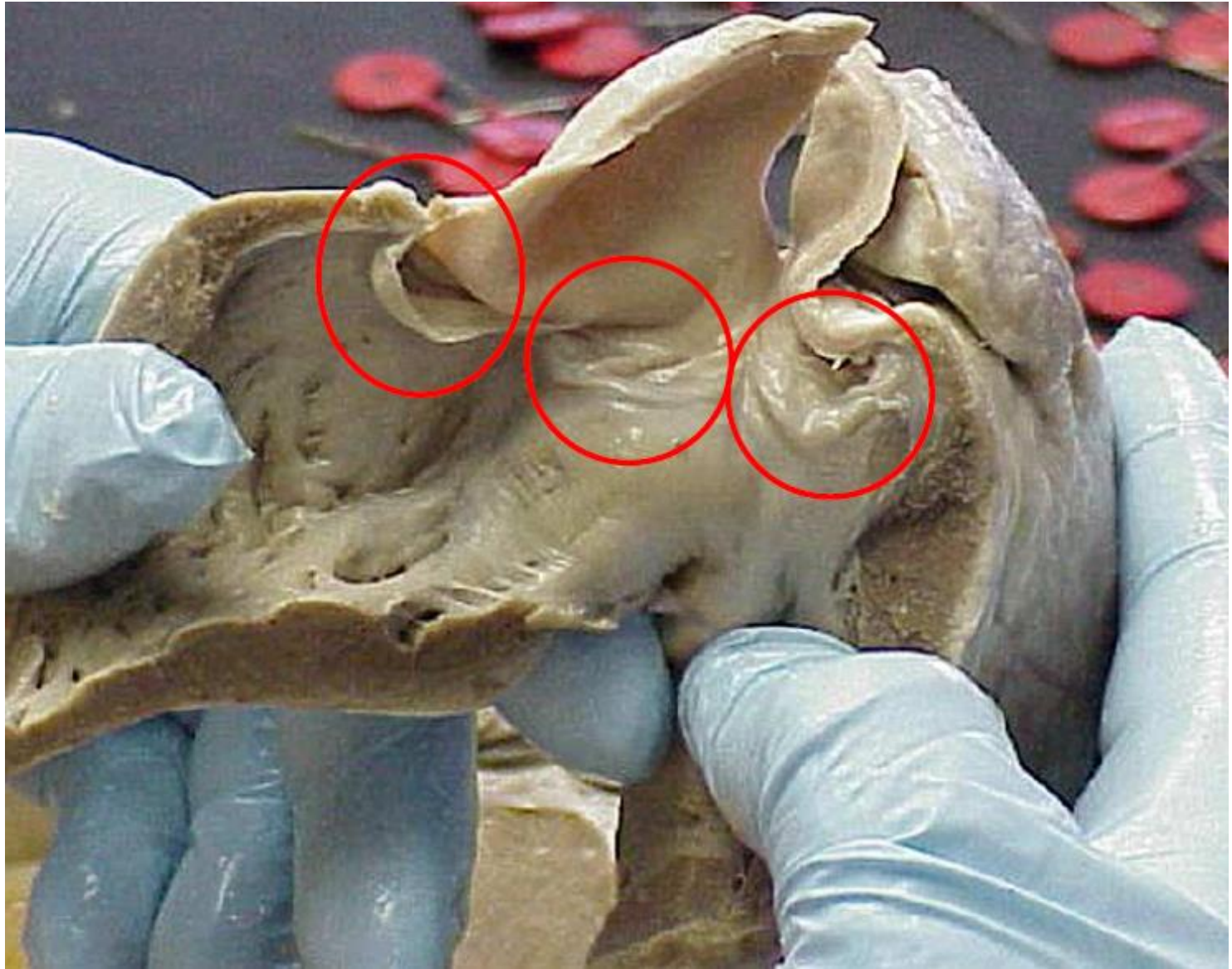


Parts of semilunar valves:
pars tensa
pars flaccida (lunula)
nodulus

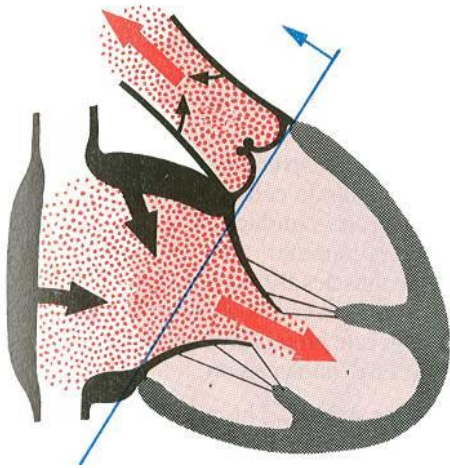


CARDIAC VALVES

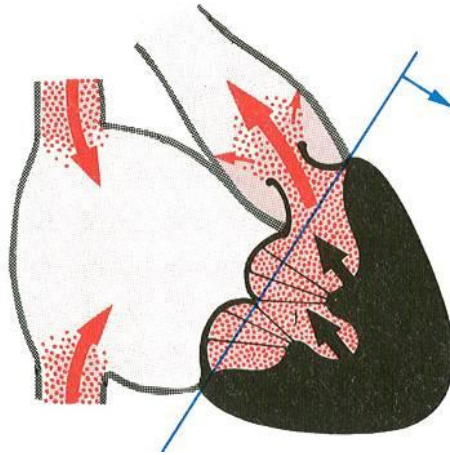
(semilunar valves/cusps - pulmonary valve)



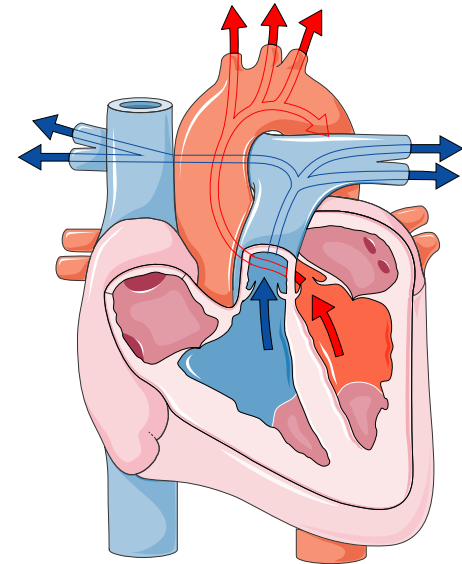
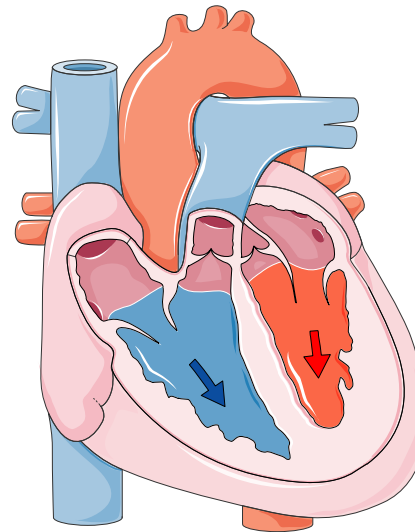
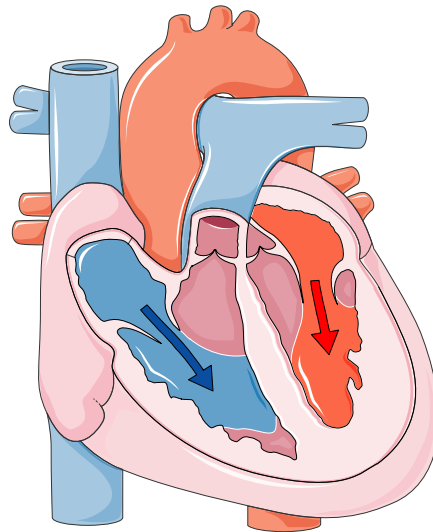
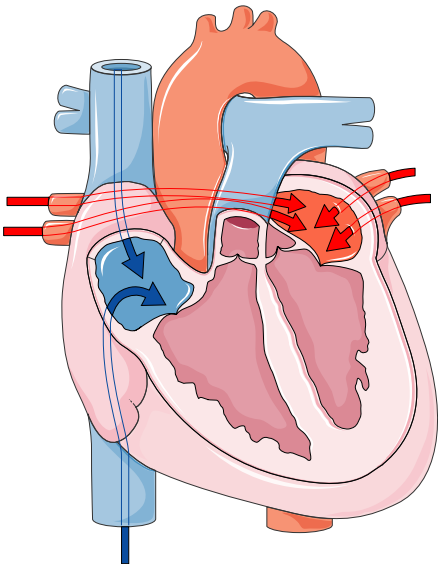
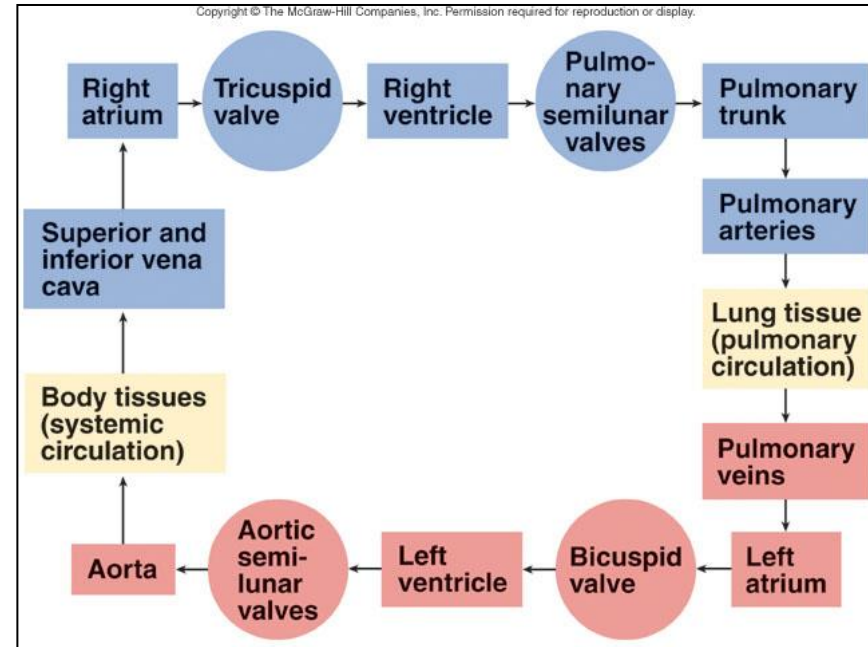
Cardiac Cycle



Diastole



Systole



Heart wall

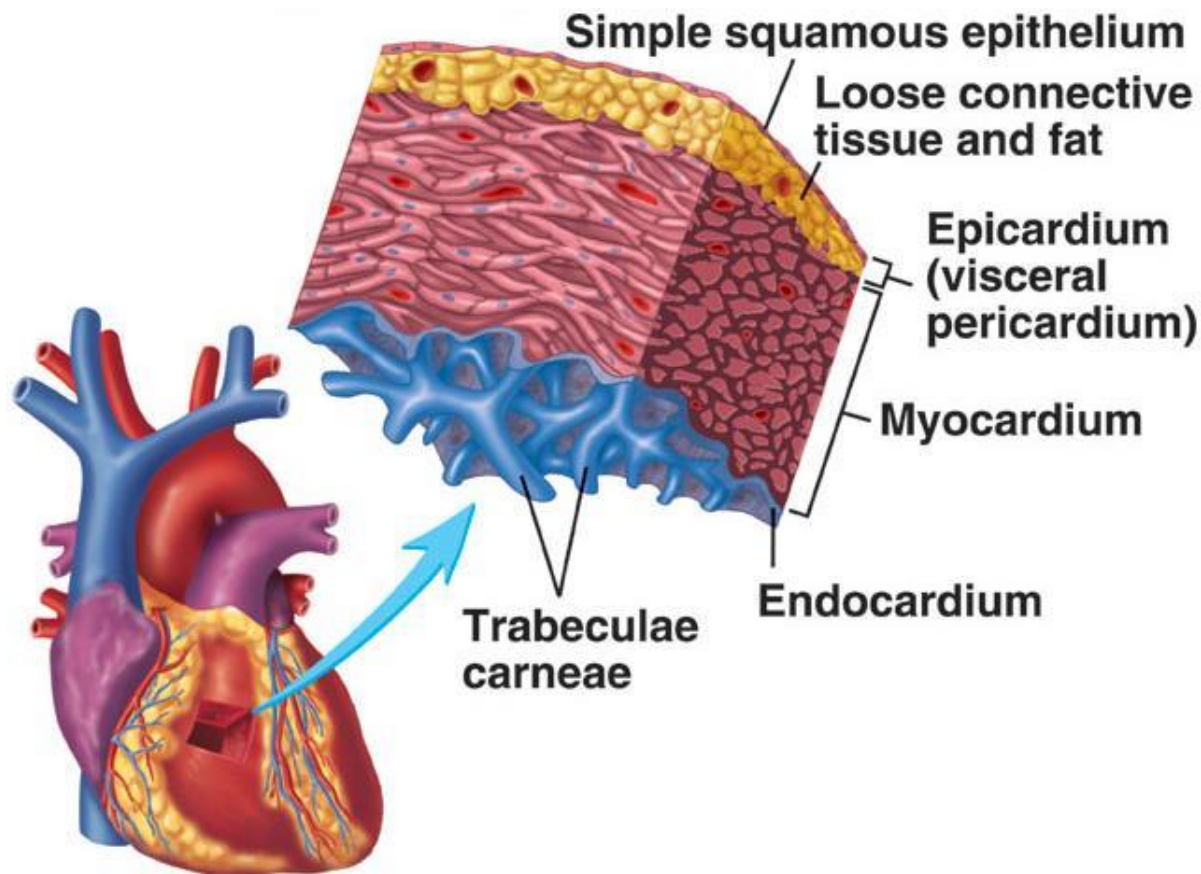
The Heart Wall

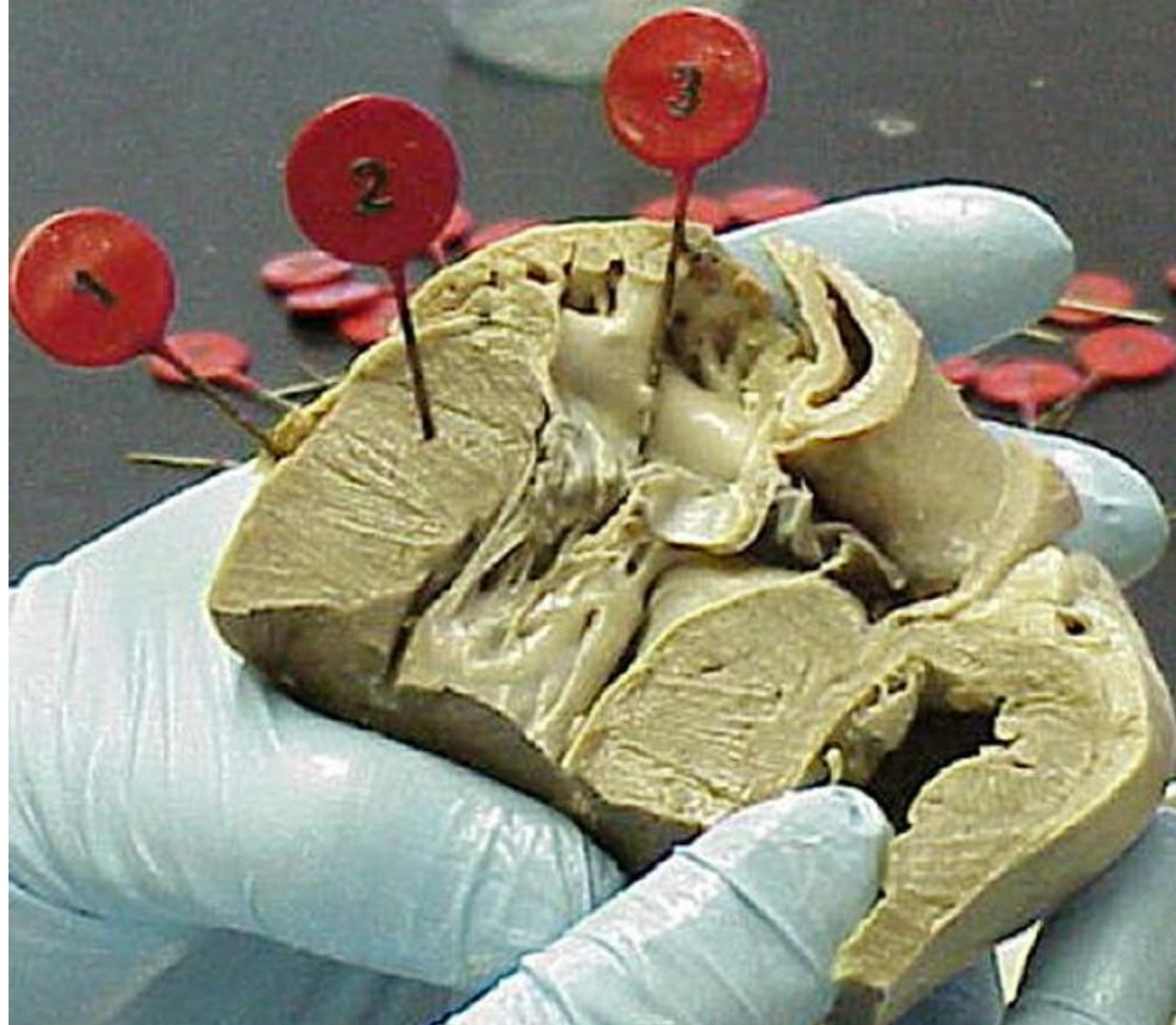
Components of the heart wall

Epicardium: Outermost serous layer

Myocardium: Middle muscle layer

Endocardium: Innermost endothelium





Myocardium

Atria: 2 layers

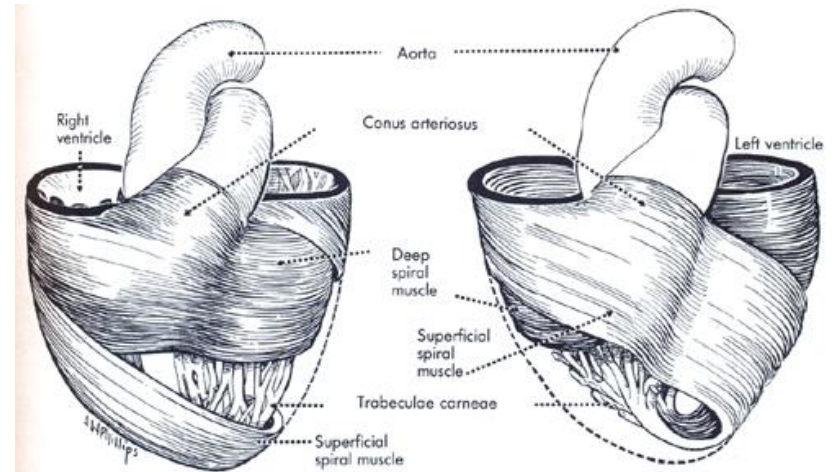
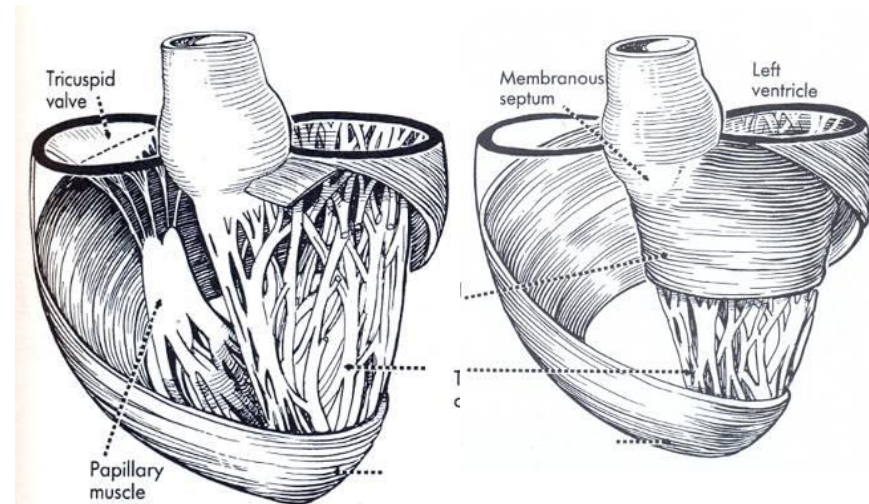
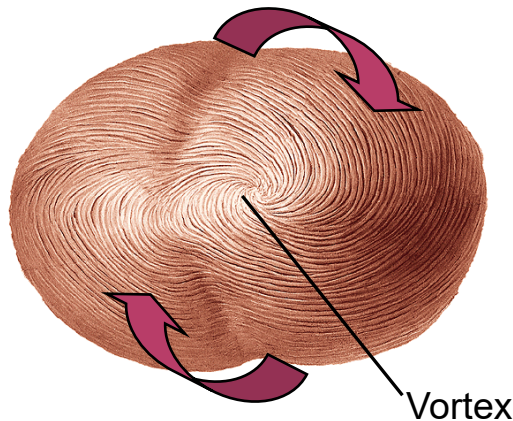
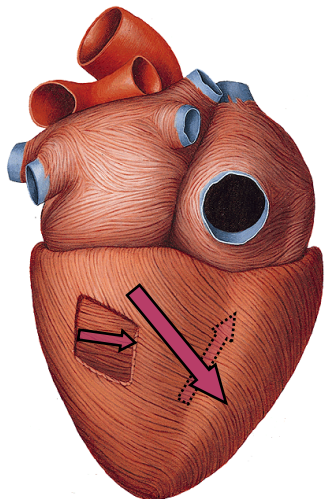
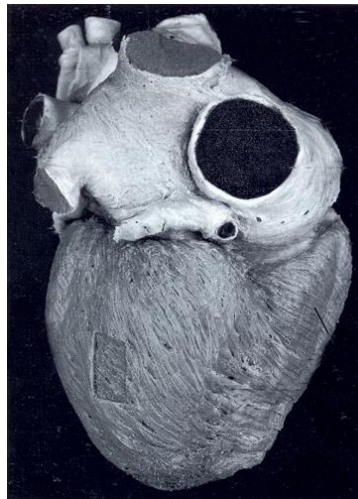
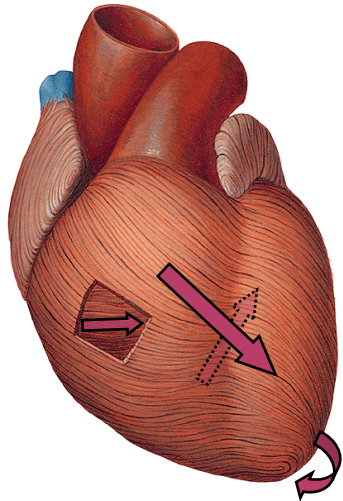
-inner longitudinal, outer circular

Ventricles: 3 layers

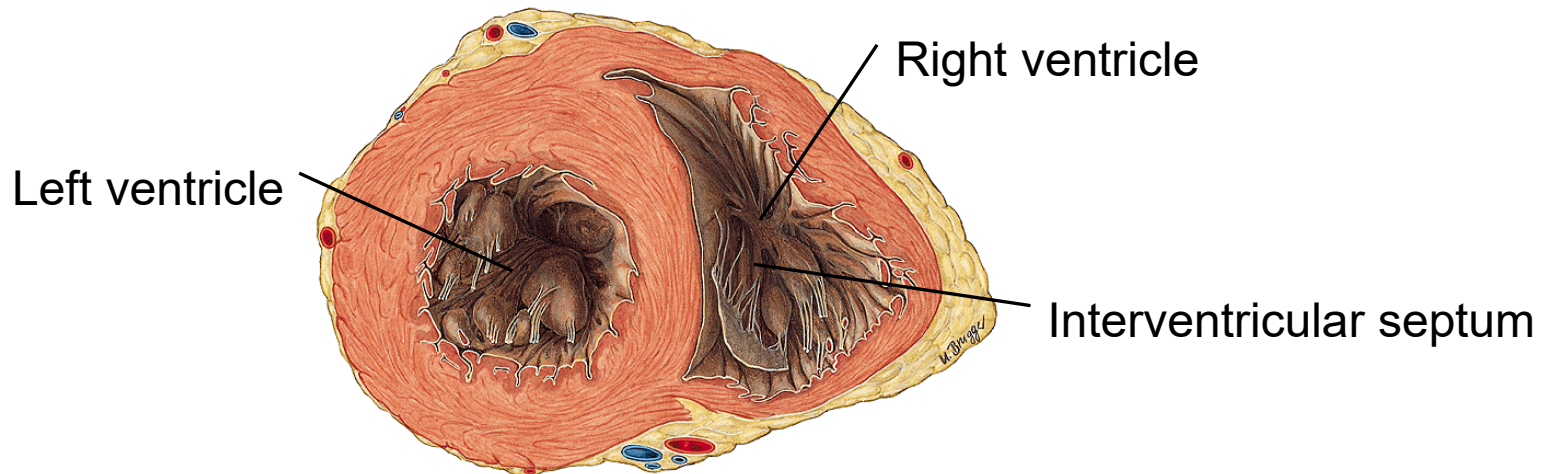
-outer spiraling longitudinal: arising from the cardiac skeleton (mainly fibrous trigones) fibers run towards the apex – they form a whorl: vortex of the heart

-inner spiraling longitudinal fibers: trabeculae carneae, papillary muscles

-middle circular: strong fibers

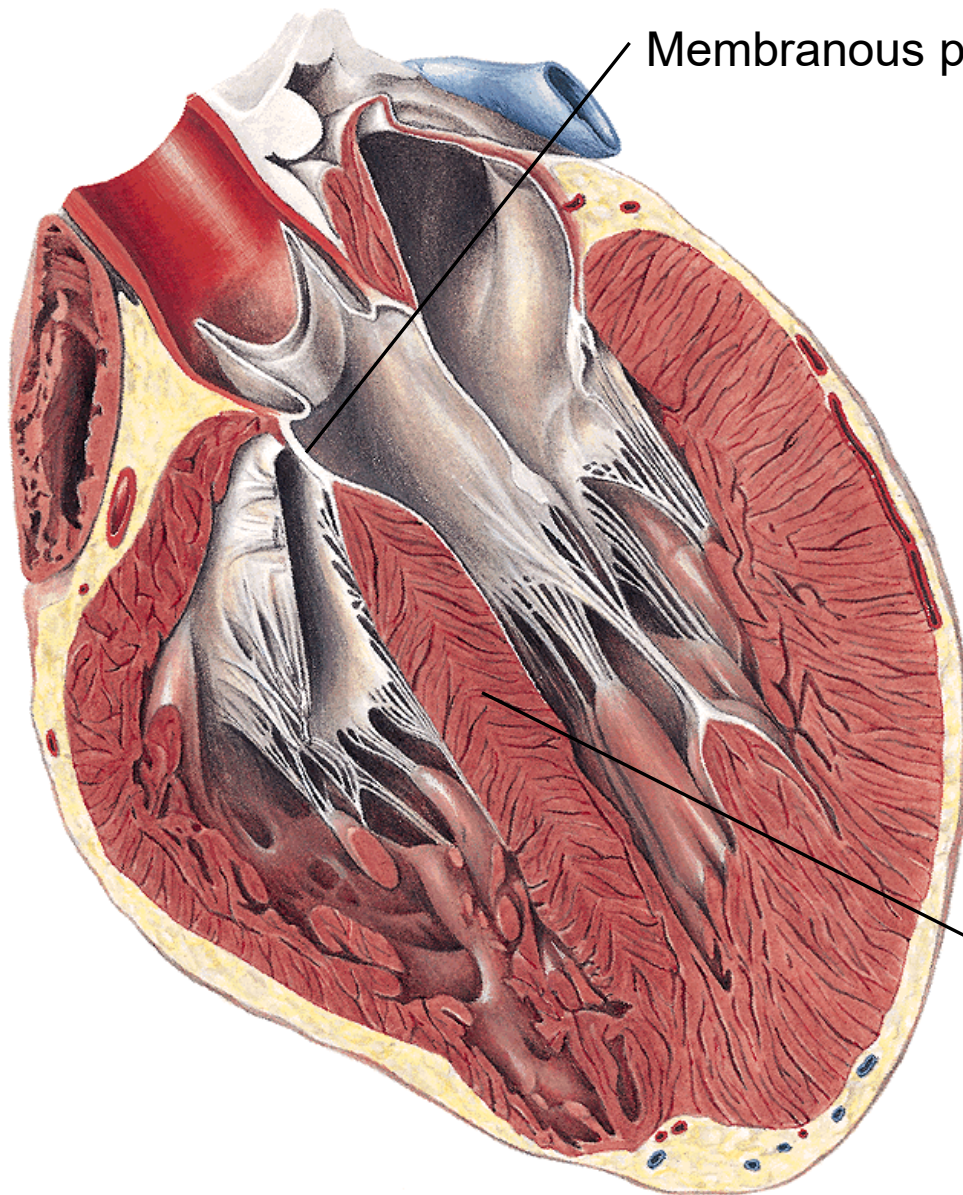


MYOCARDIUM



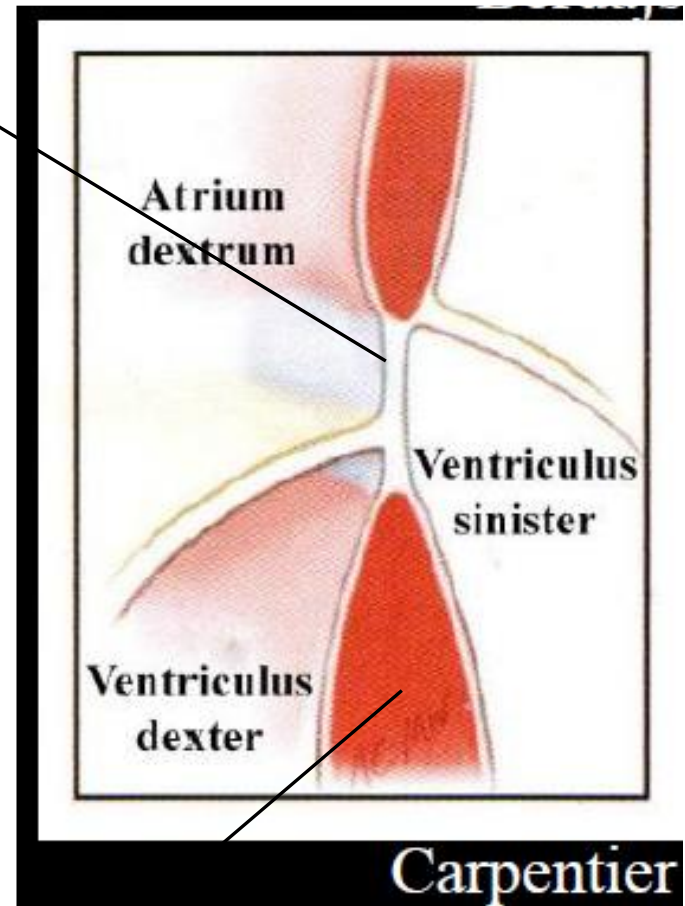
INTERVENTRICULAR SEPTUM

anterior view



Membranous part

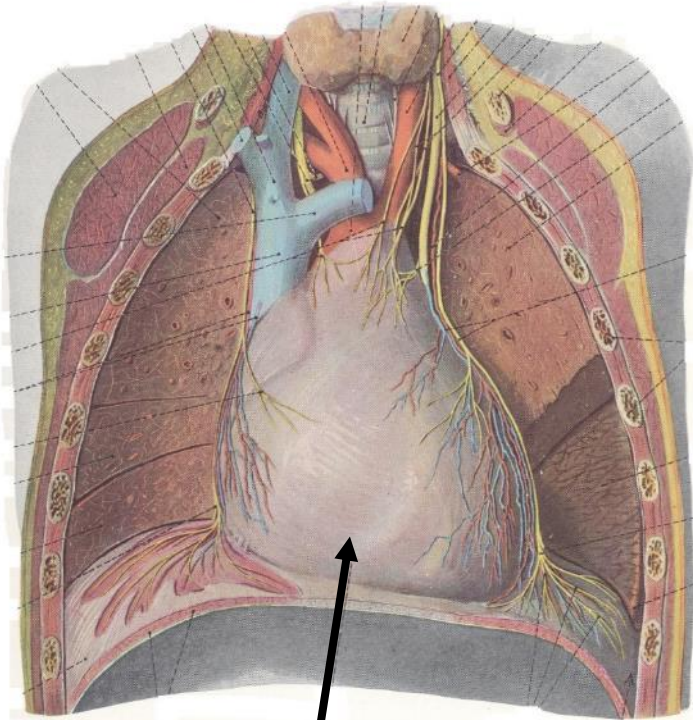
Muscular part



Pericardium

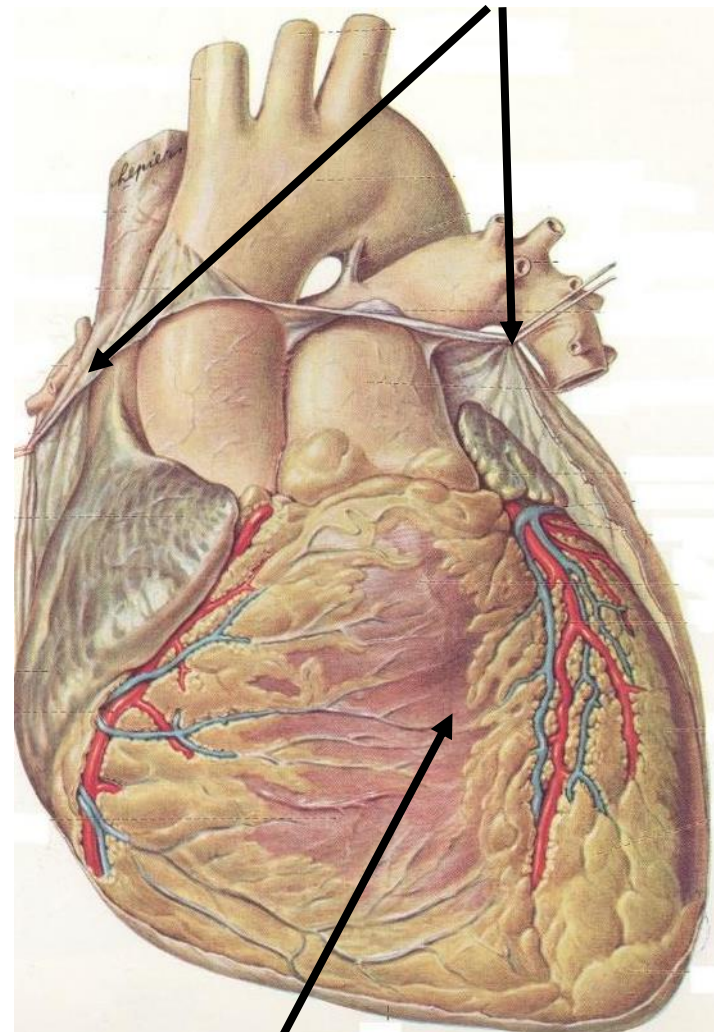
Pericardium

Layers of pericardium



fibrous pericardium

parietal layer of serous pericardium



visceral layer of serous pericardium

Pericardium

Primitive heart tube in the pericardial sac



outflow tract

epicardium (visceral pericardium)

mesocardium

pericardial cavity

fibrous pericardium

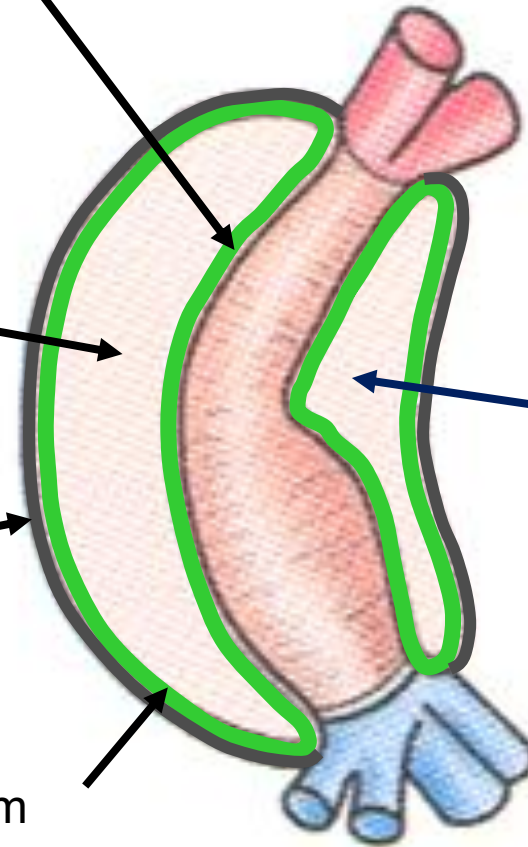
parietal pericardium

outflow tract

transverse sinus of pericardium

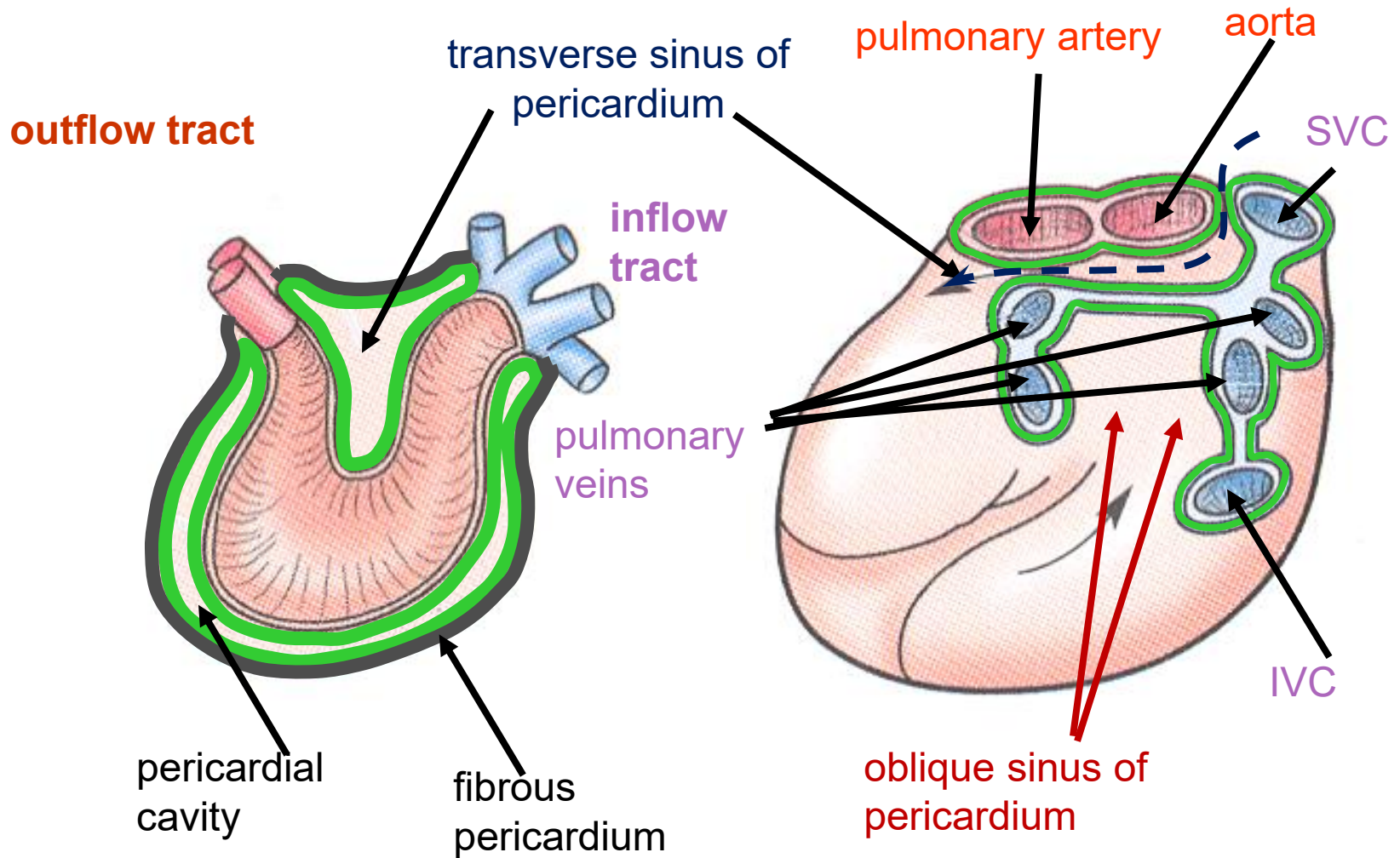
inflow tract

inflow tract



Pericardium

Primitive heart tube in the pericardial sac



**Reflection Lines
(Vein - "Sappey" - "T")**

aorta

Pericardium

pulmonary trunk

transverse sinus of
pericardium

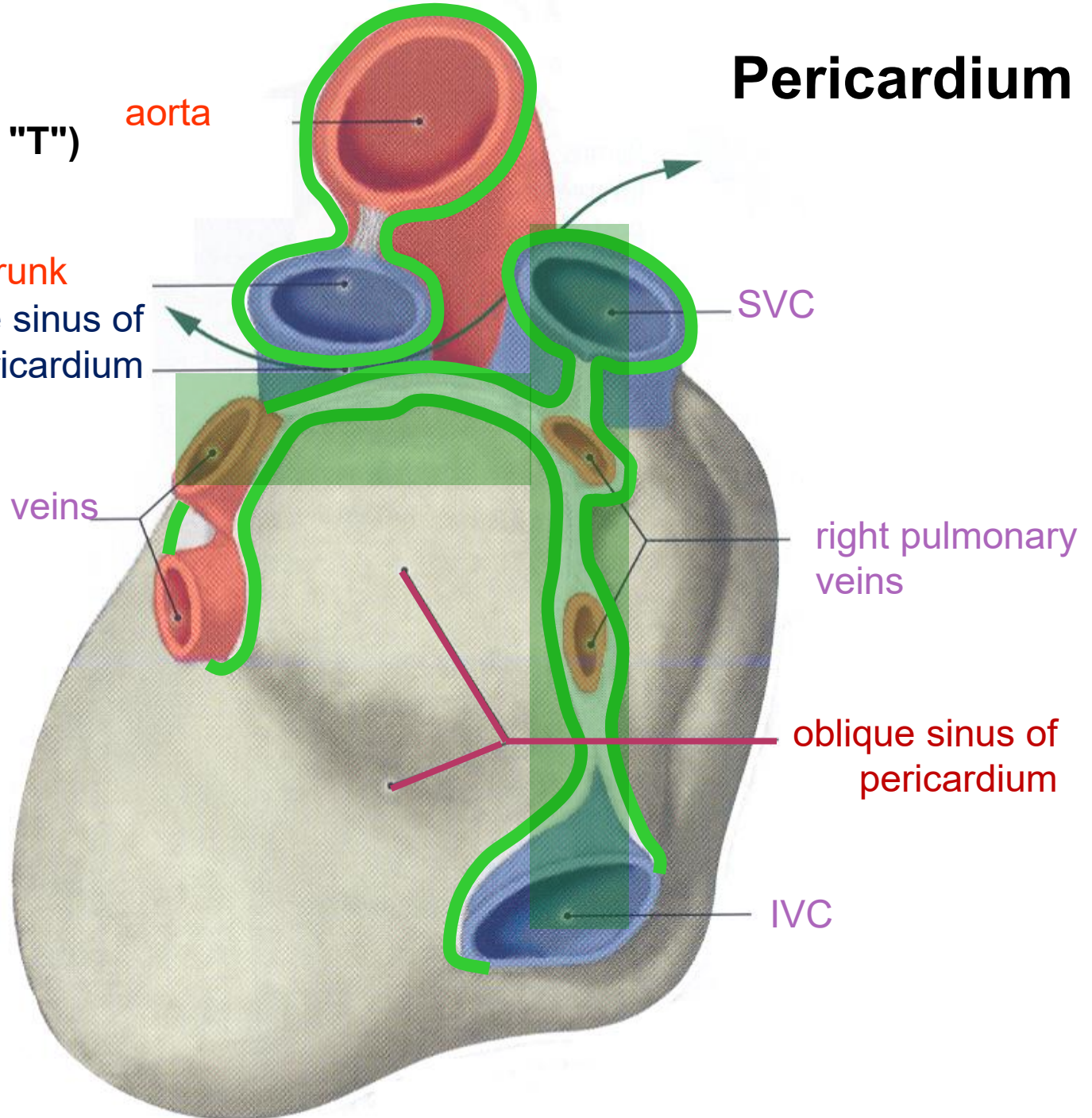
SVC

left pulmonary veins

right pulmonary
veins

oblique sinus of
pericardium

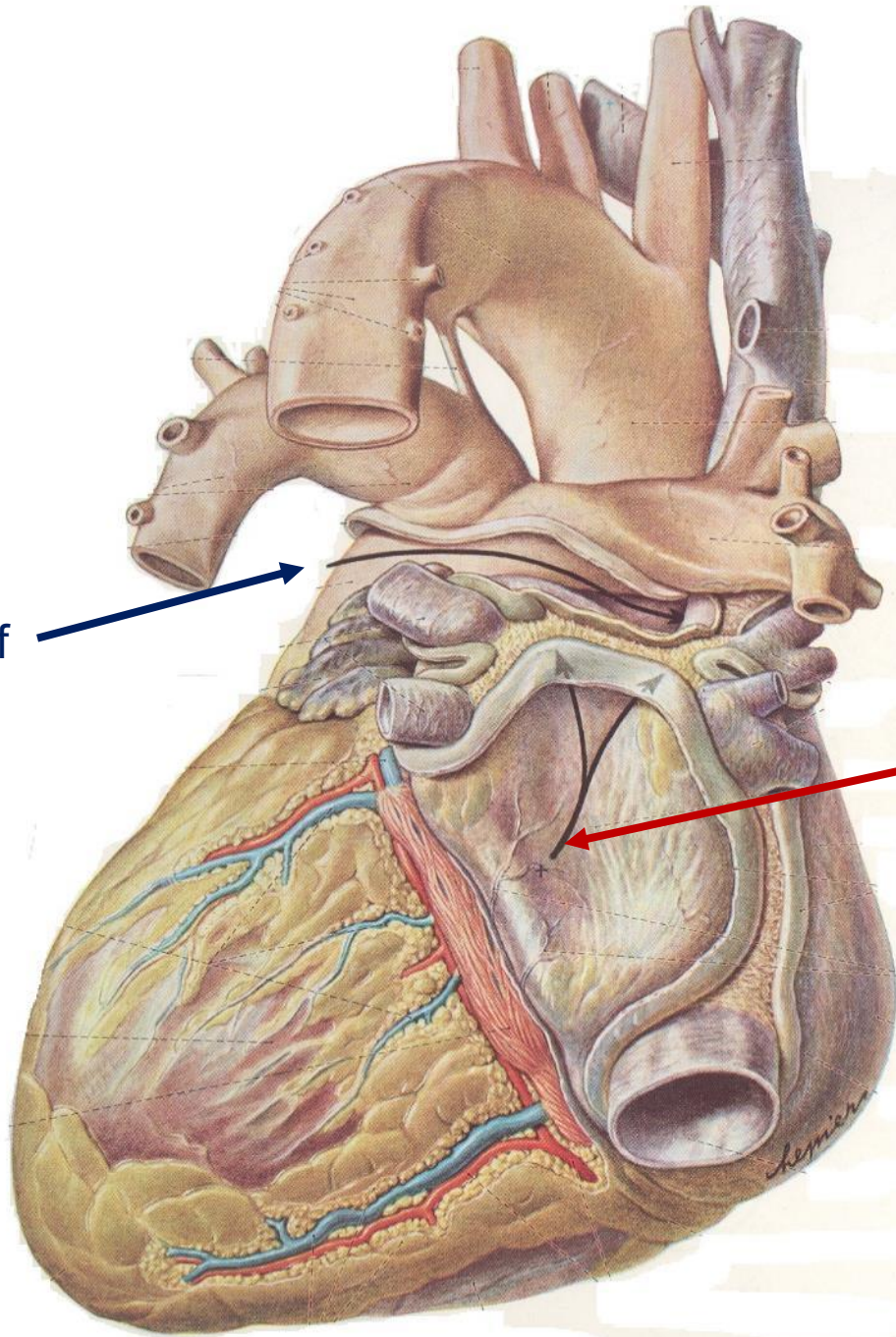
IVC



Pericardium

transverse sinus of
pericardium

oblique sinus of
pericardium



Reflection
Lines
(Vein -
"Sappey"- "T")

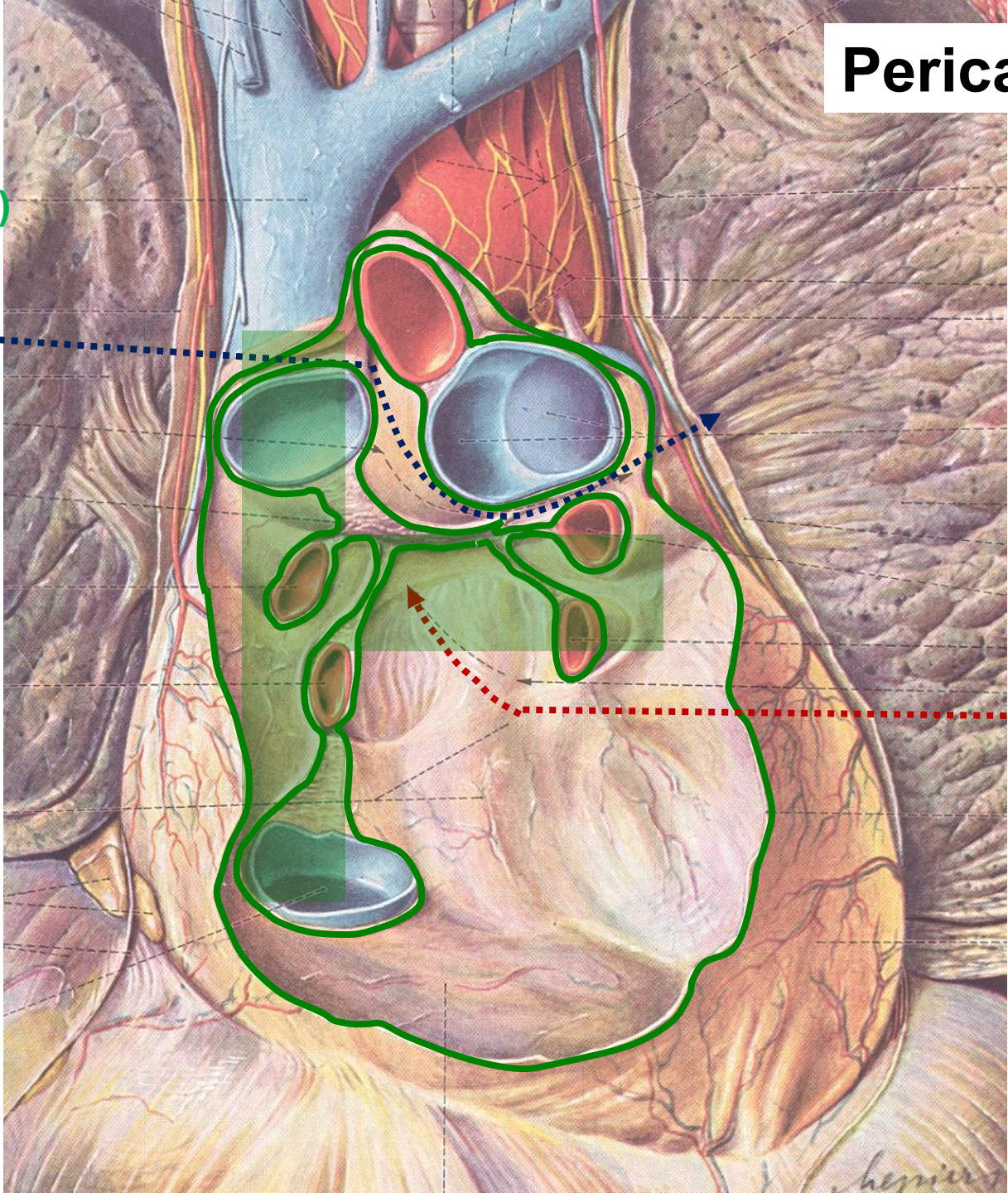
Pericardium

transverse
sinus of
pericardium

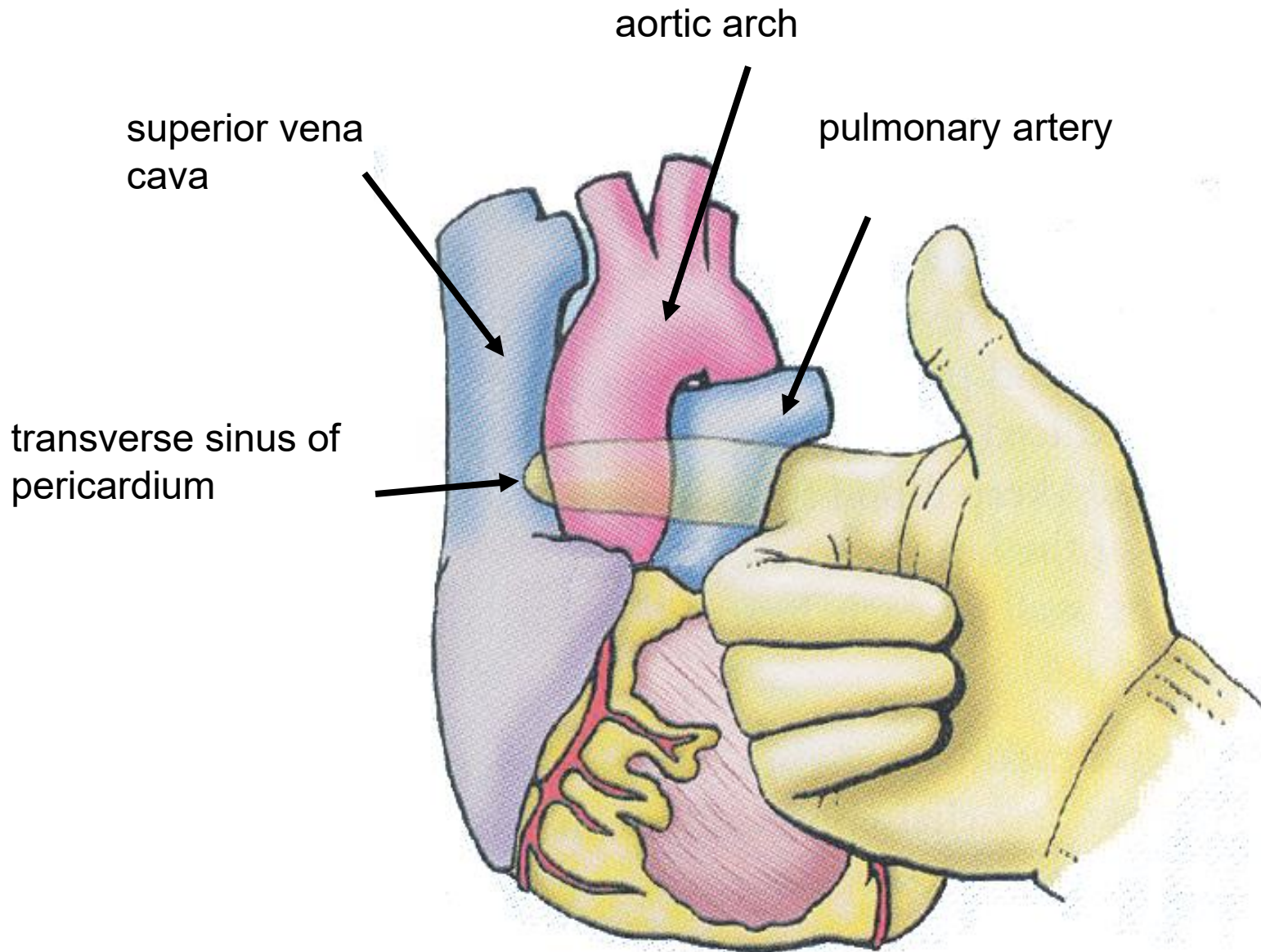
caval veins +
pulmonary
veins

aorta +
pulmonary
artery

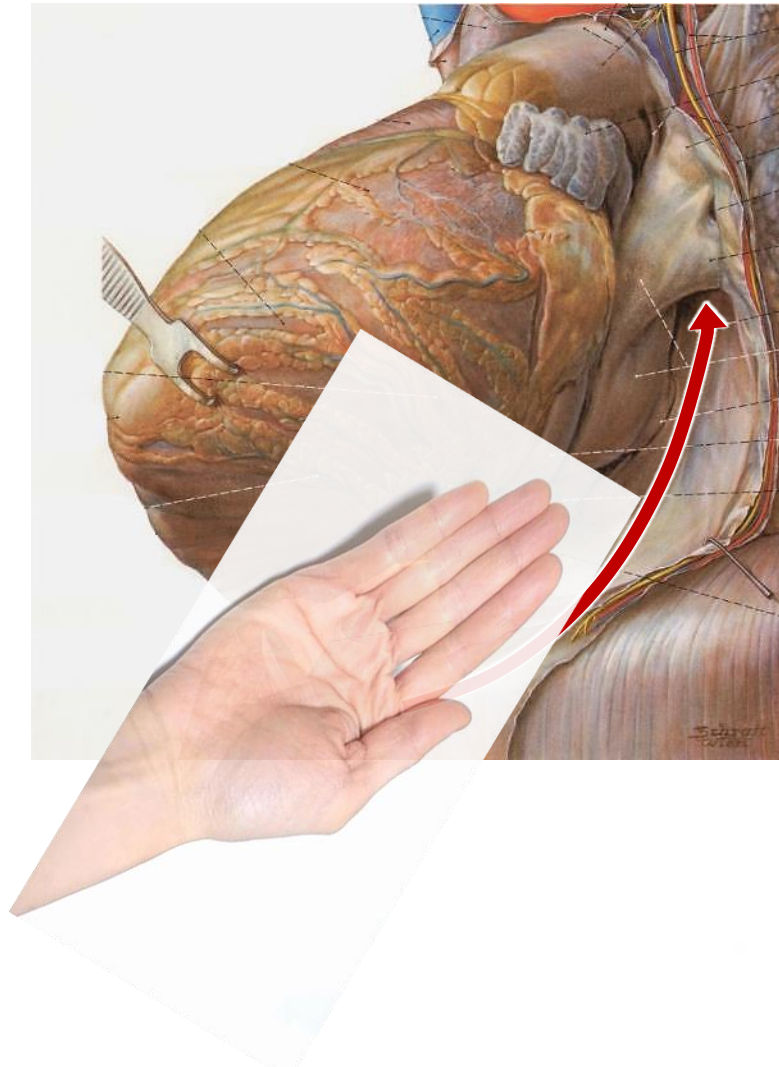
oblique
sinus of
pericardium



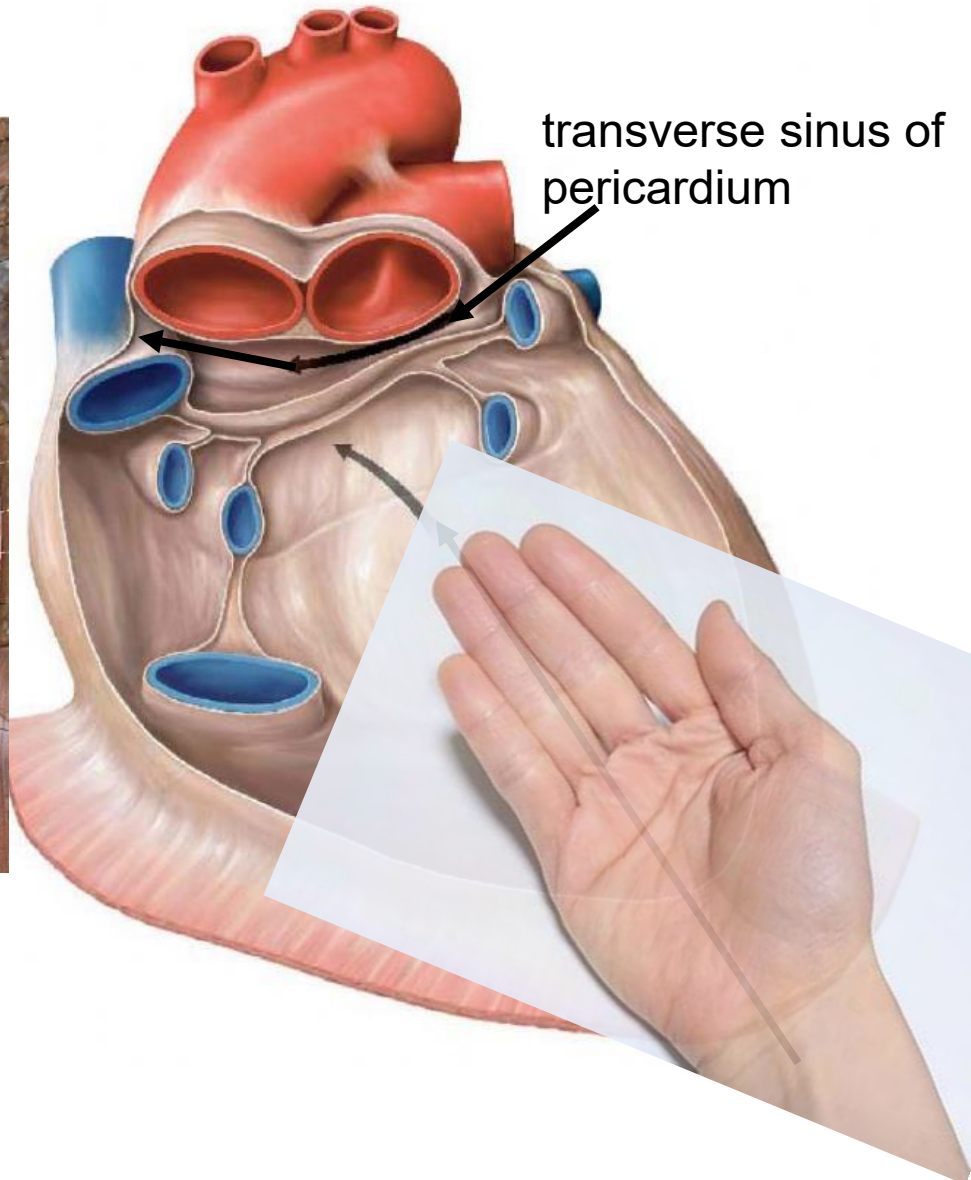
Pericardium



Pericardium



oblique sinus of pericardium



oblique sinus of pericardium

THANK YOU FOR YOUR ATTENTION!

QR CODE PLACE



REFERENCES

